Phase mixing against a passive-scalar cascade: the Hermite spectrum of kinetic reduced magnetohydrodynamic turbulence

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Abstract

In kinetic reduced magnetohydrodynamics (KRMHD) the compressive fluctuations are a kinetic field carried by the Alfvénic turbulence. Their free energy moves to fine structure in parallel velocity by phase mixing and to small perpendicular scales by the turbulence, and we ask what sets their Hermite spectrum $W(m)$. The compressive sector is exactly a continuum of passive scalars, or slices, labelled by the parallel phase velocity. Each slice is carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streams along the perturbed magnetic field at its own speed, and when the induction equation is ideal it streams along field lines that are frozen into the flow. The nonlinearity therefore moves no free energy between phase velocities and none between field-aligned wavenumbers. We propose that its only effect on $W(m)$ is to remove free energy from every Hermite moment at the same rate $1/T$, the rate at which the turbulence carries each slice to the dissipation scale, while phase mixing proceeds at the field-aligned wavenumber κ at which the free energy was injected. This gives $W(m) \propto m^{-1/2} \exp[-(2m)^{1/2}/(s\kappa T)]$, with $s = \beta_i^{1/2}$, and a break at $m_* = (s\kappa T)^2/2 \sim \beta_i/\chi^2$, where χ is the nonlinearity parameter. In forced KRMHD turbulence at $\beta_i = 1$ and $\chi \approx 1$, the perpendicular dissipation of each Hermite moment is proportional to $W(m)$ to within 20% for $m \leq 16$. With T and the distribution of κ measured independently, the model reproduces $W(m)$ to within 8% for $m \leq 30$ and the field-aligned Hermite flux to within 7% for $m \leq 24$, and neither falls below it at large m, as a return flux would require. A split of the flux by the sign of k_z reports a backward flux of about 12% of the forward flux, which the field-aligned flux does not have. At lower β_i and the same χ the break moves into the resolved range, as predicted: at $\beta_i = 0.1$, $W(m)$ falls below $m^{-1/2}$ from small m and follows the formula up to the collisional range. There the measured spectrum lies above the model, because the perpendicular sink grows with the time since the free energy was injected; against that time the sinks of all runs fall on one curve. Doubling the number of Hermite moments leaves $W(m)$ unchanged where both resolutions resolve it and halves the share of the free energy removed by hypercollisions, as the model predicts. Doubling the perpendicular resolution at $\beta_i = 0.3$ lengthens T by 40% at the same outer scale, and the field-aligned wavenumber of the compressive fluctuations grows slowly through the longer inertial range, so the uniform sink and the frozen wavenumber are approximations whose dependence on resolution is not yet settled. [*To do:* the 128^3 run at $\beta_i = 1$]

1 Introduction

Kinetic reduced magnetohydrodynamics (KRMHD), the limit $k_\perp \rho_i \to 0$ of gyrokinetics derived by Schekochihin et al. (2009), hereafter S09, describes anisotropic fluctuations at scales much larger than the ion gyroradius ρ_i in a strongly magnetised, weakly collisional plasma. Its Alfvénic fluctuations obey reduced magnetohydrodynamics (RMHD). Its compressive fluctuations, the slow

waves and the entropy mode, obey a kinetic equation for the ion distribution in the parallel velocity $v_\parallel$; they are carried by the Alfvénic fluctuations and do not act back on them. Without collisions they are damped by phase mixing (Barnes, 1966; S09, §6.2), which moves their free energy to ever finer structure in $v_\parallel$, that is, to large Hermite order m in a Hermite expansion in $v_\parallel$ (Zocco and Schekochihin, 2011; Kanekar et al., 2015; Schekochihin et al., 2016). In turbulence the same free energy is also carried to small perpendicular scales. Which of the two routes it takes decides whether the compressive cascade heats ions through Landau damping or reaches the ion gyroscale (S09; Schekochihin et al., 2016; Meyrand et al., 2019; Kawazura et al., 2019, 2020), and the Hermite spectrum $W(m)$ is the most direct record of that competition.

Two pictures of $W(m)$ in turbulence have been proposed. In the first, nonlinear advection couples fluctuations with parallel wavenumbers of opposite sign and so returns free energy from high to low m, as a stochastic plasma echo (Kanekar, 2015; Schekochihin et al., 2016; Parker et al., 2016; Adkins and Schekochihin, 2018). Meyrand et al. (2019) found in KRMHD turbulence at $\beta_i = 1$ a spectrum somewhat steeper than m^{-1}, against $m^{-1/2}$ for linear phase mixing, and a net flux to high m much smaller than the injection rate away from the forcing scales, and attributed both to such echoes. In gyrokinetic simulations Kawazura et al. (2019, 2020) found $W(m) \propto m^{-1/2}$ at high β_i and close to m^{-1} at $\beta_i = 0.1$, and read the steeper spectrum as phase mixing suppressed by echoes. In the second picture (S09, §6.3), the compressive fluctuations have no parallel cascade: the field-aligned wavenumber at which they were injected stays fixed while the Alfvén waves carry them to smaller perpendicular scales, so at small enough scales the perpendicular cascade outruns the damping. The two pictures differ on whether the compressive fluctuations acquire structure along the moving field lines. Meyrand et al. (2019) argued that they do, because resistive reconnection relabels the field lines within about one eddy-turnover time and the compressive fluctuations then inherit the parallel cascade of the Alfvénic ones.

We make the second picture quantitative and test it. Section 3 shows that the compressive sector of KRMHD is exactly a continuum of passive scalars, or slices, labelled by the parallel phase velocity u, each carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streaming along the perturbed field at its own speed. The nonlinearity therefore moves no free energy between phase velocities, and free energy moves between Hermite moments only as the correlations between slices change (section 4). When the induction equation is ideal, the field lines are frozen into the flow that carries all the slices, and every volume-integrated single-point statistic, including $W(m)$, evolves as in linear phase mixing along the initial field lines (section 5). In turbulence the slices are also cascaded to small perpendicular scales and dissipated there. Section 6 proposes that this is the only effect of the nonlinearity on $W(m)$: every Hermite moment loses free energy at the same rate $1/T$, the rate at which the turbulence carries a passive scalar to the dissipation scale, while phase mixing proceeds at the field-aligned wavenumber κ of the injected free energy. The steady state is

$$W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}} \exp\Big(-\frac{\sqrt{2m}}{s\kappa T}\Big), \qquad m_* = \frac{(s\kappa T)^2}{2} \sim \frac{\beta_i}{\chi^2}, \tag{1}$$

for injection rate ε, $s = v_{\rm th}/v_A = \beta_i^{1/2}$ and nonlinearity parameter χ, summed over the distribution of κ. Every quantity in (1) can be measured independently of $W(m)$, so the model has no free parameter. It predicts that the perpendicular dissipation of each Hermite moment is proportional to $W(m)$, that $W(m)$ and the Hermite flux computed along the field follow the model with no deficit at large m, that m_* grows in proportion to β_i at fixed χ, and that diagnostics built on the Eulerian k_z report a backward flux that the field-aligned flux does not have.

Section 7 tests these predictions in forced KRMHD turbulence with the GANDALF code (Kanekar, 2025). At $\beta_i = 1$ and $\chi \approx 1$ the perpendicular dissipation per Hermite moment is

proportional to $W(m)$ to within 20%, the model reproduces $W(m)$ to within 8% for $m \leq 30$ and the field-aligned flux to within 7% for $m \leq 24$, and the flux split by the sign of k_z has a backward part of about 12% of the forward part. At $\beta_i = 1$ the break m_* lies beyond the $M = 32$ moments we resolve. At $\beta_i = 0.3$ and 0.1, with the same Alfvénic turbulence, it moves into the resolved range as predicted, and $W(m)$ steepens accordingly. There the measured spectrum exceeds the uniform-sink model, never falls below it, and the excess identifies a refinement: the perpendicular sink grows with the age of the free energy, the same function of age at every β_i. Doubling the number of Hermite moments leaves $W(m)$ unchanged where both resolutions resolve it and halves the share of the free energy removed by hypercollisions, as the model predicts. At twice the perpendicular resolution, the perpendicular sink is 30% weaker and the field-aligned wavenumber of g grows slowly through the longer inertial range: the model's two approximations, a sink independent of the dissipation and a frozen wavenumber, hold only roughly at the resolutions we reach, while the absence of a return flux persists. [*To do:* the 128^3 run at $\beta_i = 1$]

The ingredients of (1) are not new, and we credit them where they are used. The rate comparison between Landau damping at a frozen parallel wavenumber and the perpendicular cascade is S09's (§6.3, eqs 217–220). The functional form is that of Zocco and Schekochihin (2011, eq. 112) for growing modes, with their growth rate replaced by the cascade rate, and Schekochihin et al. (2016, eqs 2.58–2.59) solved the same continuity equation in m with a collisional sink. The break scale is the threshold $k_\parallel v_{\mathrm{th}} \sim \sqrt{m}\, k_\perp u_\perp$ of Schekochihin et al. (2016, eq. 4.5), and Kawazura et al. (2019) stated $m \sim \beta_i$ for critically balanced turbulence. A decay of the Hermite spectrum as $\exp(-\lambda\sqrt{m})$, with λ proportional to the ratio of the cascade rate to $k_\parallel v_{\mathrm{th}}$, was seen in simulations of a kinetic passive scalar advected by a two-dimensional flow (Kanekar, 2015, chapter 4). What is new here is the exact statement that in three-dimensional KRMHD with ideal induction the nonlinearity moves no free energy in m, the model with independently measured inputs that follows from it, and its test in turbulence with diagnostics computed along the field. The slice decomposition itself has precedents in the G-transform of Morrison and Pfirsch (1992) and Morrison (2000), its application to drift kinetics by Heninger and Morrison (2018), and the per-velocity invariants of Plunk and Parker (2014) and Plunk (2015).

Section 2 sets out the compressive sector of KRMHD, sections 3 to 5 give the exact results, section 6 the model and its predictions, section 7 the simulations, and section 8 discusses what the results do and do not establish.

2 The compressive sector of KRMHD

For a plasma with one ion species and isothermal electrons, KRMHD (Schekochihin et al., 2009) consists of reduced magnetohydrodynamics (RMHD) for the Alfvénic fluctuations and a kinetic equation for the compressive ones. We use the form of CMM (Chandran et al., 2026), whose section 3.2 and appendix B restate and extend S09; their equation numbers refer to arXiv:2607.27981v1. Table 3 in appendix A compares our conventions with those of CMM, S09 and the GANDALF code (Kanekar, 2025).

2.1 Alfvénic fields

We set the Alfvén speed $v_A = 1$ unless stated otherwise and take the mean field $\boldsymbol{B}_0$ along $\hat{\boldsymbol{z}}$. The stream function Φ and the flux function Ψ give the $\boldsymbol{E} \times \boldsymbol{B}$ velocity $\boldsymbol{u}_E = \hat{\boldsymbol{z}} \times \nabla_\perp \Phi$ and the perpendicular magnetic fluctuation $\boldsymbol{b}_\perp \equiv \delta\boldsymbol{B}_\perp/(4\pi n_i m_i)^{1/2} = \hat{\boldsymbol{z}} \times \nabla_\perp \Psi$, and to first order the unit vector along the field is $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \boldsymbol{b}_\perp$; both $\boldsymbol{u}_E$ and $\hat{\boldsymbol{b}}$ are divergence-free. With the bracket

$\{f,h\} = \partial_x f\,\partial_y h - \partial_y f\,\partial_x h = \hat{\boldsymbol{z}}\cdot(\nabla f\times\nabla h)$, advection by $\boldsymbol{u}_E$ is $\{\Phi,\cdot\}$ and the derivative along the field is $\nabla_\parallel \equiv \hat{\boldsymbol{b}}\cdot\nabla = \partial_z + \{\Psi,\cdot\}$. The RMHD equations are (Schekochihin et al., 2009, eqs 17–18)

$$\partial_t\Psi + \{\Phi,\Psi\} = \partial_z\Phi, \qquad \partial_t\nabla_\perp^2\Phi + \{\Phi,\nabla_\perp^2\Phi\} = \nabla_\parallel\nabla_\perp^2\Psi. \tag{2}$$

The first, the induction equation $\partial_t\Psi = \nabla_\parallel\Phi$, states that the electric field along the total magnetic field vanishes. Equations (2) are CMM's eqs (3.23)–(3.24) in their normalised potentials (3.2) and do not contain the compressive fluctuations. The Elsasser potentials $z^\pm = \Phi \pm \Psi$ (S09's $\zeta^\pm$) obey $\partial_t z^\pm = \pm\partial_z z^\pm$ linearly, so z^+ propagates toward $-z$ and z^- toward $+z$.[1]

2.2 Kinetic equation and the two branches

The ion thermal speed is $v_{\rm th} = (2T_i/m_i)^{1/2}$, so $s \equiv v_{\rm th}/v_A = \beta_i^{1/2}$; we write $\tau = T_i/T_e$ and $Z = q_i/e$. The compressive part of the ion distribution is the function g of S09 (CMM's g_{09}, their eq. 3.26), which obeys CMM's eq. (3.25) (S09 eq. 157 without collisions),

$$\frac{\mathrm{d}g}{\mathrm{d}t} + v_\parallel\nabla_\parallel\Big[g + \Big(\frac{Z}{\tau}\frac{\delta n}{n} + \frac{v_\perp^2}{v_{\rm th}^2}\frac{\delta B}{B}\Big)F_0\Big] = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t} = \partial_t + \{\Phi,\cdot\}, \tag{3}$$

where F_0 is the three-dimensional Maxwellian. Quasineutrality and perpendicular pressure balance give (CMM eqs 3.27–3.28; S09 eqs 158–159),

$$\overline{g} = \frac{\delta n}{n} - \frac{\delta B}{B}, \qquad \overline{(v_\perp^2/v_{\rm th}^2)\,g} = -\frac{Z}{\tau}\frac{\delta n}{n} - 2\Big(1+\frac{1}{\beta_i}\Big)\frac{\delta B}{B}, \qquad \overline{(\cdots)} \equiv \frac{1}{n_i}\int \mathrm{d}^3v\,(\cdots). \tag{4}$$

These relations are algebraic: $\delta n/n$ and $\delta B/B$ at a point are linear functionals of g at that point, with coefficients that do not depend on the wavenumber. At finite $k_\perp\rho_i$ they contain gyroaveraging operators.

We expand g in Laguerre polynomials of $v_\perp^2/v_{\rm th}^2$ with coefficients $G_l(v_\parallel)$ (S09 eq. 205; CMM eq. B8; appendix A). The source term in (3) is linear in $v_\perp^2$, so the coefficients with $l \geq 2$ obey $\mathrm{d}G_l/\mathrm{d}t + v_\parallel\nabla_\parallel G_l = 0$ (CMM eq. B10; S09 eqs 210 and 212). The coefficients with $l = 0, 1$ are coupled through $\delta n/n$ and $\delta B/B$; the combinations $G^\pm$ of CMM's eqs (B5)–(B6) (S09 eqs 183–184) obey CMM's eq. (B4) (S09 eq. 181),

$$\frac{\mathrm{d}G^\pm}{\mathrm{d}t} + v_\parallel\nabla_\parallel G^\pm = \frac{v_\parallel F_M(v_\parallel)}{\Lambda^\pm}\,\nabla_\parallel\int G^\pm(v_\parallel')\,\mathrm{d}v_\parallel', \tag{5}$$

where $F_M(v_\parallel) = \pi^{-1/2}v_{\rm th}^{-1}e^{-v_\parallel^2/v_{\rm th}^2}$ and

$$\Lambda^\pm = -\frac{\tau}{Z} + \frac{1}{\beta_i} \pm \Delta, \qquad \Delta = \Big[\Big(1+\frac{\tau}{Z}\Big)^2 + \frac{1}{\beta_i^2}\Big]^{1/2} \tag{6}$$

(CMM eq. B7; S09 eqs 182 and 204).[2] We call the right-hand side of (5), of strength $1/\Lambda^\pm$, the Landau coupling; in linear theory it produces the collisionless damping of the compressive fluctuations (S09, §6.2.2). The coefficients with $l \geq 2$ obey (5) with $1/\Lambda = 0$, and we call them the sector $\Lambda = \infty$. The compressive sector therefore consists of independent kinetic equations for G^+, G^- and each G_l with $l \geq 2$, all advected by the same Alfvénic fields. Since $\Delta > 1 + \tau/Z$ and $\Delta > 1/\beta_i$, $\Lambda^+ - 1 > \beta_i^{-1}$ and $\Lambda^- < -\tau/Z$, so $\Lambda^+ > 1$ and $\Lambda^- < 0$ for all $\beta_i, \tau/Z > 0$ (S09, below eq. 200); at $\beta_i = \tau = Z = 1$, $\Lambda^\pm = \pm\sqrt{5}$.

[1] S09's §§2.3 and 2.5 write the solutions with $\zeta^\mp = 0$ as $\zeta^\pm = f^\pm(x, y, z \mp v_A t)$, the opposite direction. We checked ours against S09's eq. (21), CMM's equations and GANDALF's right-hand side.

[2] CMM's eq. (B16) prints the square root in (6) as $[(1+\tau/Z) + \beta_i^{-2}]^{1/2}$, without the square on $1+\tau/Z$; with the square, as in S09's eq. (204), their decompositions (B15) and (B17) hold, and without it they do not (appendix A).

2.3 One branch in the variable u

We now drop the branch label, set $u = v_\parallel/v_{\rm th}$, and write $F_M(u) = \pi^{-1/2}e^{-u^2}$ and $M_k[G] = \int u^k G\,\mathrm{d}u$. Since $v_\parallel = su$ when $v_A = 1$, equation (5) becomes

$$\partial_t G + \{\Phi, G\} + s\,\nabla_\parallel(LG) = 0, \tag{7}$$

with the branch operator

$$LG = uG - \frac{uF_M}{\Lambda}\,M_0[G], \tag{8}$$

which acts on the velocity variable only; for $\Lambda = \infty$ it is multiplication by u. Because the field relations (4) are algebraic, L contains no wavenumber.

For given Φ and Ψ, equation (7) is linear in G. Its nonlinear terms are the advection $\{\Phi, G\}$ and the part $s\{\Psi, LG\}$ of the streaming, which follows the perturbed field lines. The Alfvénic fields obey (2) whatever G does, so the compressive sector is passive. At finite $k_\perp\rho_i$ this fails; the induction equation, for example, contains $\nabla_\parallel(\delta n/n)$ (CMM, §3.1).

The density and field-strength fluctuations are linear combinations of $M_0[G^+]$ and $M_0[G^-]$ (A3), the parallel ion flow involves $M_1[G^\pm]$, and the pressure perturbations involve $M_0[G^\pm]$ and $M_2[G^\pm]$ (appendix A); the sector $\Lambda = \infty$ contributes to none of these moments.

2.4 Hermite representation

Following CMM (eqs B27 and B30), we expand $G = \sum_{m\geq 0} g_m H_m(u) F_M(u)/\sqrt{2^m m!}$, so that $g_m = \int H_m\, G\,\mathrm{d}u/\sqrt{2^m m!}$ and $G/F_M^{1/2} = \sum_m g_m \psi_m(u)$ with the orthonormal Hermite functions ψ_m. The g_m are CMM's $\tilde{G}_m$, and $M_0[G] = g_0$, $M_1[G] = g_1/\sqrt{2}$. In this basis (7) is the hierarchy

$$\partial_t g + \{\Phi, g\} + s\,\nabla_\parallel(Ag) = 0, \qquad g = (g_0, g_1, g_2, \dots)^{\mathsf{T}}, \tag{9}$$

where the ladder matrix A, the matrix of L, is tridiagonal with zero diagonal,

$$A_{m,m+1} = A_{m+1,m} = \sqrt{\frac{m+1}{2}} \quad \text{except} \quad A_{10} = \frac{1 - 1/\Lambda}{\sqrt{2}}. \tag{10}$$

The weight matrix $\mathsf{P} = \mathrm{diag}(p_m)$, with $p_0 = 1 - 1/\Lambda$ and $p_m = 1$ for $m \geq 1$, symmetrises it, $\mathsf{P}A = (\mathsf{P}A)^{\mathsf{T}}$, and $\sum_m p_m g_m^2$ is the free-energy product $[G, G]$ of section 3. A truncation at $m = M$ needs a closure; by default we use the zero closure $g_{M+1} = 0$, which keeps the first $M+1$ rows and columns of A. GANDALF (Kanekar, 2025) integrates (2) in Elsasser form together with (9) for one branch, with optional dissipation, hypercollisions and forcing, truncated with the zero closure or with the copy closure $g_{M+1} = g_{M-1}$ (section 4.4). We checked to round-off that its right-hand side without dissipation and hypercollisions is (9)–(10), for both signs of Λ and both closures.

The rest of the paper concerns one equation (7) with $\Lambda \notin [0, 1]$, which we call a branch whether it describes $G^\pm$ or a Laguerre coefficient with $l \geq 2$. Sections 3 to 5 treat a collisionless and unforced branch except where stated; sections 6 and 7 add forcing, dissipation and hypercollisions. The Hermite spectrum of the free energy is $W(m) = p_m\langle g_m^2\rangle$, where $\langle\cdot\rangle$ is the average over the periodic box, so that $\sum_m W(m) = \langle[G, G]\rangle$.

3 Phase-velocity slices

This section shows that each branch (7) is a continuum of passive scalars, or slices, labelled by the parallel phase velocity. The Alfvénic fields Φ and Ψ are given functions of $\boldsymbol{r}$ and t; (2) is not used.

3.1 Without the Landau coupling

For $\Lambda = \infty$ the operator L is multiplication by u, and (7) holds separately at every u:

$$\partial_t G + \{\Phi, G\} + su\,\nabla_\parallel G = 0.$$

Since $\nabla_\parallel = \partial_z + \{\Psi, \cdot\}$ and $\{\chi, G\} = (\hat{\boldsymbol{z}} \times \nabla_\perp \chi) \cdot \nabla G$, this is the advection equation $\partial_t G + \boldsymbol{U}_u \cdot \nabla G = 0$ with the velocity

$$\boldsymbol{U}_u = \hat{\boldsymbol{z}} \times \nabla_\perp (\Phi + su\Psi) + su\,\hat{\boldsymbol{z}} = \boldsymbol{u}_E + su\,\hat{\boldsymbol{b}}, \qquad \nabla \cdot \boldsymbol{U}_u = 0. \tag{11}$$

At each u, G is a passive scalar, carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streaming along the perturbed field at the speed su, which is the motion of the guiding centre of a free ion with $v_\parallel = uv_{\mathrm{th}}$. Values of G at different u do not interact. This is the exact equation for the Laguerre coefficients G_l with $l \geq 2$ (S09, eqs 210 and 212).

3.2 With the Landau coupling

For finite Λ, $LG = uG - (uF_M/\Lambda)\,M_0[G]$ ties every velocity to the density $M_0[G]$, through the parallel electric field and the mirror force, so G at fixed u is no longer a passive scalar. Two properties of L restore the picture. First, L acts on velocity only and is the same at every point and at every wavenumber, because the field relations (4) are algebraic. Second, L is self-adjoint in the free-energy product

$$[f, g] = \int \frac{fg}{F_M}\,\mathrm{d}u - \frac{M_0[f]\,M_0[g]}{\Lambda}, \tag{12}$$

which is positive definite for $\Lambda \notin [0, 1]$, as for every compressive branch of KRMHD ($\Lambda^+ > 1$, $\Lambda^- < 0$; section 2). It therefore has a complete set of generalised eigenfunctions, the Case–Van Kampen modes F_u of the branch (van Kampen, 1955; Case, 1959), with $LF_u = uF_u$ for every real u. Each is a beam of particles at $v_\parallel = uv_{\mathrm{th}}$ dressed by the response of the Maxwellian, through the Landau coupling, to the density of the beam. Expanding G in these modes is a fixed linear change of variables in velocity, $G \mapsto a_u$, with

$$a_u = \frac{\varepsilon_R\,G - \varepsilon_I\,\mathsf{H}[G]}{\sqrt{F_M}\,|\varepsilon|}(u), \tag{13}$$

where $\varepsilon(u + \mathrm{i}0) = \varepsilon_R + \mathrm{i}\varepsilon_I$ is the dielectric function of the branch on the real axis, which contains the plasma dispersion function, and H is the Hilbert transform in u (appendix B, (B3)). For $\Lambda = \infty$, $\varepsilon = 1$ and $a_u = G/\sqrt{F_M}$. The change of variables is invertible, divides the free energy among phase velocities without cross terms, and turns L into multiplication by u:

$$[G, G] = \int a_u^2\,\mathrm{d}u, \qquad G \mapsto a_u \;\text{ takes }\; LG \mapsto u\,a_u \tag{14}$$

(theorem 4). Its inverse is, up to normalisation, the G-transform that Morrison and Pfirsch (1992) and Morrison (2000) introduced for linearised Vlasov–Poisson; Heninger and Morrison (2018) applied it to electrostatic drift kinetics, where the transformed components remain coupled through the potential. The function $\varrho = F_M/|\varepsilon|^2$ that appears below is the spectral density of the branch, with $\int \varrho\,\mathrm{d}u = 1/p_0$ and $\varrho = F_M$ for $\Lambda = \infty$.

3.3 The slice theorem

Theorem 1. *Assume that (i) $\Lambda \notin [0,1]$, the value $\Lambda = \infty$ included; (ii) $\Phi(\boldsymbol{r},t)$ and $\Psi(\boldsymbol{r},t)$ are real functions on $\mathbb{T}^3 \times [0,t_1]$, with $\mathbb{T}^3$ a periodic box, whose gradients are continuous and Lipschitz in $\boldsymbol{r}$, uniformly in t; they need not satisfy (2) and may obey resistive or forced RMHD, or be prescribed; (iii) G and uG are continuously differentiable functions of $(\boldsymbol{r},t)$ with values in the space $\mathcal{W}$ of functions of u with finite $\int G^2 F_M^{-1}\,\mathrm{d}u$ (appendix B). Then G solves (7) if and only if, at every $(\boldsymbol{r},t)$ and for almost every u,*

$$\partial_t a_u + \boldsymbol{U}_u \cdot \nabla a_u = \partial_t a_u + \{\Phi + su\Psi,\, a_u\} + su\,\partial_z a_u = 0, \tag{15}$$

with $\boldsymbol{U}_u$ of (11).

Proof. The map $\mathcal{A} : G \mapsto a$ of (13) is bounded and injective (theorem 4) and acts on u only, through a kernel independent of $\boldsymbol{r}$ and t, so it commutes with ∂_t, with spatial derivatives and with multiplication by functions of $(\boldsymbol{r},t)$, hence with $\{\Phi,\cdot\}$, $\{\Psi,\cdot\}$ and $\nabla_\parallel$. With $\mathcal{A}LG = u\,\mathcal{A}G$ this gives $\mathcal{A}[\partial_t G + \{\Phi,G\} + s\,\nabla_\parallel(LG)] = \partial_t a + \{\Phi,a\} + su\,\nabla_\parallel a$, and $u\,\nabla_\parallel a_u = u\,\partial_z a_u + \{u\Psi, a_u\}$ gives (15). □

With the Landau coupling, therefore, each slice is again a passive scalar carried by $\boldsymbol{U}_u$, in three dimensions, and slices with different u do not interact; the Landau coupling has been absorbed into the definition of the slices. The parallel electric field and the mirror force, which enter through the Landau coupling, only dress the slices: the distribution at fixed u is not a passive scalar, but its projection a_u on F_u is. In terms of the Elsasser potentials, the stream function of the slice at u is

$$\Phi + su\Psi = \tfrac{1}{2}\big[(1+su)\,z^+ + (1-su)\,z^-\big], \tag{16}$$

so the slices at $su = -1$ and $+1$ are advected by z^- alone and by z^+ alone, and the slice at $u = 0$, the distribution at $v_\parallel = 0$, by Φ alone. The sectors have different transforms, but $\boldsymbol{U}_u$ depends only on u, so at each u the slices of G^+, G^- and every G_l with $l \geq 2$ are carried by the same flow:

$$(\partial_t + \boldsymbol{U}_u \cdot \nabla)(a_u^+,\, a_u^-,\, a_u^{(2)},\, a_u^{(3)}, \dots) = 0. \tag{17}$$

The compressive sector of KRMHD is therefore a continuum of passive scalars labelled by phase velocity.

3.4 Dissipation, collisions and forcing

Let the right-hand side of (7) be $-\mathcal{B}G - \mathcal{C}G + F$, where the linear operators $\mathcal{B}$ and $\mathcal{C}$ act on $\boldsymbol{r}$ alone and on u alone, respectively, and F is a source. The proof of theorem 1 gives

$$\partial_t a_u + \boldsymbol{U}_u \cdot \nabla a_u = -\mathcal{B}\,a_u - \int \hat{\mathcal{C}}(u,u')\,a_{u'}\,\mathrm{d}u' + \sqrt{\varrho(u)}\;\mathcal{T}[F](u), \tag{18}$$

where $\hat{\mathcal{C}}$ is the kernel of $\mathcal{A}\,\mathcal{C}\,\mathcal{A}^{-1}$. Damping at a rate that depends only on $k_\perp$ or k_z, applied with the same coefficient to every Hermite moment, is of the form $\mathcal{B}$ and damps each slice separately. With $e_m = H_m(u)F_M(u)/\sqrt{2^m m!}$ and $r_m(u)$ the mth Hermite coefficient of F_u, a polynomial of degree m with $r_0 = 1$ and $r_1 = \sqrt{2}\,u$, one has $\mathcal{T}[e_m](u) = p_m\,r_m(u)$, and the functions $\phi_m = \sqrt{p_m \varrho}\;r_m$ form an orthonormal basis of $L^2(\mathbb{R},\mathrm{d}u)$. For $\mathcal{C}e_m = \nu_m e_m$, as for hypercollisions $\nu_m = \nu(m/M)^n$, the kernel is $\hat{\mathcal{C}}(u,u') = \sum_m \nu_m\,\phi_m(u)\,\phi_m(u')$, which couples different slices unless all ν_m are equal; collisions are the only term in (18) that exchanges free energy between slices. Forcing of g_0 by a field $f(\boldsymbol{r},t)$, $F = f\,e_0$, gives every slice the same spatial pattern, with amplitude $(1-1/\Lambda)\sqrt{\varrho(u)}\,f$.

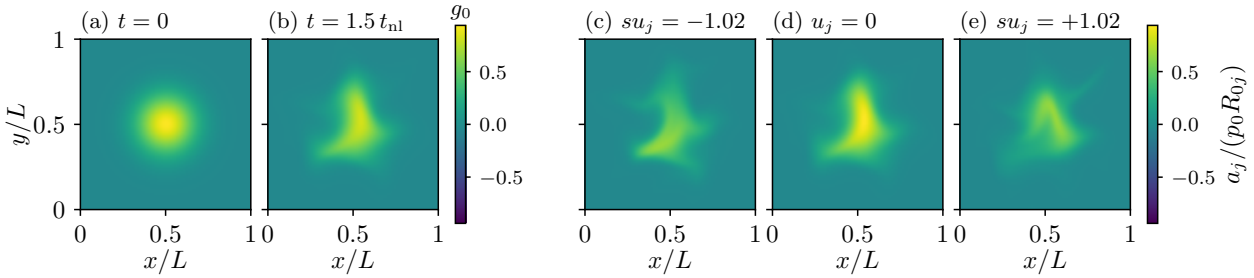


Figure 1: A density blob in decaying Alfvénic turbulence, on the plane $z = 0$ (coordinates in units of the box side $L = 1$). GANDALF run on a 64^3 grid with $M = 16$ (zero closure), $\Lambda = \sqrt{5}$ and $\beta_i = 1$; balanced Gaussian random Elsasser fields with $1 \le k_\perp/k_1 \le 3$, $|k_z|/k_1 \le 2$ and $\chi = 1.00$; hyper-resistivity damps $z^\pm$ and every g_m; no forcing or hypercollisions; $g(0) = f\,e_0$, with f a Gaussian of standard deviation $1/8$ in x and y and $1/4$ in z; t runs to $1.5\,t_{\rm nl}$, with $t_{\rm nl} = 0.1357$ the outer eddy time. (a, b) The density moment g_0 at $t = 0$ and $1.5\,t_{\rm nl}$. (c–e) The slices $a_j/(p_0 R_{0j})$ of section 4.4 at $t = 1.5\,t_{\rm nl}$ at the nodes $su_j = -1.0218$, $u_j = 0$ and $su_j = +1.0218$; all three equal f at $t = 0$. Each group of panels has one symmetric colour scale.

3.5 Slices in a turbulent run

Figure 1 releases a density blob, $g(0) = f\,e_0$, into decaying Alfvénic turbulence with $\chi = 1.00$. Every slice starts with the pattern f and is then carried by its own stream function $\Phi + su_j\Psi$, which at $su_j = -1.0218$, 0 and $+1.0218$ is mainly z^-, Φ alone and mainly z^+; the three slices end visibly different, and g_0 mixes them. GANDALF's $\partial_t g$ projected on the slices agrees with (26), computed with brackets that do not use GANDALF, to 5.2×10^{-16} of the largest slice rate, while advecting every slice by Φ alone gives relative errors of 0.30 to 1.46. Over the run each slice energy changes only by the hyper-resistive loss of that slice, to 1.2×10^{-6} of its initial energy, so no free energy passes between slices.

4 Free energy and the Hermite spectrum in slice variables

This section collects the consequences of theorem 1 for the free energy and its Hermite spectrum. Without collisions, dissipation or sources the energy of every slice is conserved, so the free energy is not moved between phase velocities, and free energy moves between Hermite moments only as the correlations between slices change.

4.1 Slice energies

Let $\mathcal{E}(u) = \langle a_u^2 \rangle$ be the slice energy, so that $\frac{1}{2} n_i T_i\, \mathcal{E}(u)\,\mathrm{d}u$ is the free energy per unit volume in the slices with phase velocities in $[u, u + \mathrm{d}u]$ and $\int \mathcal{E}\,\mathrm{d}u = \langle [G, G] \rangle = \sum_m W(m)$ by (14). Under the hypotheses of theorem 1, with no collisions, dissipation or sources acting on G, (11) and $\nabla \cdot \boldsymbol{U}_u = 0$ give $\partial_t a_u^2 + \nabla \cdot (\boldsymbol{U}_u\, a_u^2) = 0$, so $\mathcal{E}(u)$ is independent of time for every u. Every functional $\int \varphi(u)\,\mathcal{E}(u)\,\mathrm{d}u$ is then conserved. The free energy $W^\pm$ of each branch (S09) and the two further quadratic invariants found by Chandran et al. (2026), $\Gamma^\pm$ and the phase-space helicity $\mathcal{H}^\pm$ (their eqs B25 and B18), are the members $\varphi = 1$, u and $1/u$, up to normalisation; we do not use the last two here.

The nonlinearity of KRMHD therefore moves no free energy between phase velocities. Of the terms that act on G in a simulation, damping at a rate that depends only on $k_\perp$ or k_z, applied

alike to every Hermite moment, removes free energy from each slice separately (18), and only velocity-dependent collisions exchange it between slices.

4.2 The two-slice law and the Hermite flux

Let $C(u,u') = \langle a_u\, a_{u'}\rangle$ be the two-slice correlation, whose diagonal is $\mathcal{E}(u)$.

Proposition 1 (Two-slice law)**.** *Under the hypotheses of theorem 1, with no collision operator, source or dissipation acting on G, and for slice amplitudes that are continuously differentiable in $\boldsymbol{r}$ and t with derivatives continuous in u,*

$$\frac{\mathrm{d}C(u,u')}{\mathrm{d}t} = -s\,(u-u')\,\langle a_{u'}\,\nabla_\parallel a_u\rangle. \tag{19}$$

Proof. By (15), $\partial_t a_u = -\{\Phi, a_u\} - su\,\nabla_\parallel a_u$, and the Leibniz rule gives $\partial_t(a_u a_{u'}) = -\{\Phi, a_u a_{u'}\} - su'\,\nabla_\parallel(a_u a_{u'}) - s\,(u-u')\,a_{u'}\nabla_\parallel a_u$. Since $\nabla\cdot\boldsymbol{u}_E = \nabla\cdot\hat{\boldsymbol{b}} = 0$, the first two terms are divergences, whose box averages vanish. □

The $\boldsymbol{E}\times\boldsymbol{B}$ flow carries the two slices of a pair together and cannot change their correlation, which changes only through their relative streaming along the perturbed field. Phase mixing and unmixing are therefore the decorrelation and recorrelation of slices along $\nabla_\parallel$, a definition that refers to no wavenumber.

Free energy moves between Hermite moments only through the streaming term. In the Hermite form (9) the bracket does not change $\langle g_m^2\rangle$, and since $\mathsf{P}A$ is symmetric and $\langle g_n\nabla_\parallel g_{n'}\rangle$ antisymmetric, only the coupling between m and $m+1$ survives in $E_{\le m} = \sum_{n\le m} W(n)$:

$$\frac{\mathrm{d}E_{\le m}}{\mathrm{d}t} = -\Pi_{m+1/2}, \qquad \Pi_{m+1/2} = s\,b_m\langle g_m\nabla_\parallel g_{m+1} - g_{m+1}\nabla_\parallel g_m\rangle, \qquad b_m = p_m A_{m,m+1}, \tag{20}$$

the free energy per unit time and volume that passes from moment m to $m+1$. In slice variables $\Pi_{m+1/2} = -\iint K_m(u,u')\,\partial_t C(u,u')\,\mathrm{d}u\,\mathrm{d}u'$, with the Christoffel–Darboux kernel $K_m(u,u') = \sum_{n\le m}\phi_n(u)\,\phi_n(u')$ of the functions ϕ_n of section 4.3, which by (19) is a weighted integral of the pair correlations $\langle a_{u'}\nabla_\parallel a_u\rangle$. The flux must be computed with the full $\nabla_\parallel$: the Hermite flux of Meyrand et al. (2019) uses it, whereas a flux computed with ∂_z alone, which we call the k_z-only flux, omits the part of the streaming that follows the tilt of the field lines and does not close the budget below.

With forcing of g_0 at the rate ε, dissipation that acts on every g_m in the same way and hypercollisions, the free energy in each moment obeys

$$\frac{\mathrm{d}W(m)}{\mathrm{d}t} = \Pi_{m-1/2} - \Pi_{m+1/2} + \varepsilon\,\delta_{m0} - D_\perp(m) - D_\nu(m), \qquad \Pi_{-1/2} = 0, \tag{21}$$

where $D_\perp(m)$ and $D_\nu(m)$ are the rates at which the spatial dissipation and the hypercollisions remove free energy from moment m. The brackets and the hyper-resistivity of GANDALF act on all moments alike, so (21) holds for its equations up to the error of the dealiased brackets, with the flux (20) evaluated with the same brackets; in the simulations of section 7 it closes to $4\times10^{-4}\,\varepsilon$.

4.3 The Hermite spectrum and the two-slice correlation

With the orthonormal basis $\phi_m = \sqrt{p_m\varrho}\,r_m$ of $L^2(\mathbb{R},\mathrm{d}u)$ (section 3.4), $g_m = p_m^{-1/2}\int\phi_m\,a_u\,\mathrm{d}u$, and the Hermite spectrum is a quadratic form in the two-slice correlation,

$$W(m) = \iint C(u,u')\,\phi_m(u)\,\phi_m(u')\,\mathrm{d}u\,\mathrm{d}u'. \tag{22}$$

If the slices are completely decorrelated, $C(u,u') = e(u)\,\delta(u-u')$, then $W(m) = \int e\,\phi_m^2\,\mathrm{d}u$. At $\Lambda = \infty$, ϕ_m is the Hermite function ψ_m, and for $e = F_M$

$$\int F_M\,\psi_m^2\,\mathrm{d}u = \frac{\Gamma(m+\frac{1}{2})}{\sqrt{2}\,\pi\,m!} = \frac{1}{\pi\sqrt{2m}}\Big(1 - \frac{1}{8m} + O(m^{-2})\Big), \tag{23}$$

where $\Gamma(\cdot)$ is Euler's gamma function. This is the $m^{-1/2}$ spectrum of linear phase mixing. For one Fourier mode whose density moment is driven by white noise, the exact steady state of Kanekar et al. (2015, eq. 4.10) is the spectrum of decorrelated slices with $e = \pi(1-1/\Lambda)^2\epsilon_k\,\varrho/(s|k_z|)$; at $\Lambda = \infty$

$$W(m) = \frac{\pi\epsilon_k}{s|k_z|}\,\frac{\Gamma(m+\frac{1}{2})}{\sqrt{2}\,\pi\,m!} \;\simeq\; \frac{\varepsilon}{s|k_z|\sqrt{2m}} \quad (m \gg 1), \tag{24}$$

and at every Λ the large-m form is $\varepsilon/(s|k_z|\sqrt{2m})$, with ε the free-energy injection rate, and the flux (20) is ε through every m. Density forcing loads every slice with the same spatial pattern (section 3.4), two slices loaded together dephase at the rate $s|k_z(u-u')|$, and the steady state superposes all past injections, so the slices are decorrelated in the mean. Free energy injected at $m = 0$ thus moves up the ladder at the speed $\mathrm{d}m/\mathrm{d}t = \Pi/W = s|k_z|\sqrt{2m}$ and reaches $m \simeq (sk_zt)^2/2$ after a time t.

Correlated slices give steeper spectra. If $C(u,u') = [e(u)e(u')]^{1/2}\,c(u-u')$ at $\Lambda = \infty$, with e slowly varying, the large-m form of ψ_m gives

$$W(m) \simeq \frac{\hat{c}(\sqrt{2m})}{\pi\sqrt{2m}}\int e\,\mathrm{d}u, \qquad \hat{c}(q) = \int c(\delta)\cos(q\delta)\,\mathrm{d}\delta, \tag{25}$$

so $\sqrt{2m}$ acts as a wavenumber in velocity space (Schekochihin et al., 2016; Adkins and Schekochihin, 2018). A power law $\hat{c} \propto q^{-p}$ gives $W \propto m^{-(p+1)/2}$, and a Lorentzian correlation $c(\delta) = \delta_c/[\pi(\delta^2+\delta_c^2)]$ of width δ_c gives $\hat{c} = e^{-q\delta_c}$ and $W(m) \propto m^{-1/2}\exp(-\sqrt{2m}\,\delta_c)$. A spectrum that falls exponentially in $\sqrt{m}$ is therefore one in which slices further apart than δ_c in phase velocity are decorrelated and nearer ones are not.

4.4 Truncated hierarchies

Codes such as GANDALF evolve the moments $g_0,\dots,g_M$ of a branch and close the hierarchy at $m = M$. For the zero closure A is the leading $(M+1)\times(M+1)$ block of the ladder matrix (10); the copy closure $g_{M+1} = g_{M-1}$ changes, for $M \geq 2$, only $A_{M,M-1}$, to $\sqrt{M/2} + \sqrt{(M+1)/2}$.

Theorem 2. *Let $\Lambda \notin [0,1]$, the value $\Lambda = \infty$ included, and use the zero closure or, for $M \geq 2$, the copy closure.*

(i) A has $M+1$ real and simple eigenvalues $u_0 < \cdots < u_M$. The right eigenvector of u_j is $\boldsymbol{r}_j = (r_0(u_j),\dots,r_M(u_j))^{\mathsf{T}}$, with r_m the polynomials of section 3.4; the left eigenvector is $\boldsymbol{w}_j = \mathsf{P}\boldsymbol{r}_j$, and $\boldsymbol{w}_j^{\mathsf{T}}\boldsymbol{r}_l = n_j\,\delta_{jl}$ with $n_j > 0$.

(ii) For arbitrary real Φ and Ψ, and brackets evaluated by any bilinear rule, possibly followed by linear operations, such as dealiasing, that act on every g_m alike, g solves (9) if and only if the slice amplitudes $a_j = \boldsymbol{w}_j^{\mathsf{T}}g/\sqrt{n_j} = \sum_m p_m R_{mj} g_m$, with $R_{mj} = r_m(u_j)/\sqrt{n_j}$, satisfy

$$\partial_t a_j + \{\Phi + su_j\Psi,\, a_j\} + su_j\,\partial_z a_j = 0, \qquad j = 0,\dots,M. \tag{26}$$

(iii) $g_m = \sum_j R_{mj}\,a_j$, and $\sum_m p_m|g_m(\boldsymbol{k})|^2 = \sum_j |a_j(\boldsymbol{k})|^2$ for every Fourier mode.

Proposition 2. *For the zero closure, the u_j are the zeros of r_{M+1}, and with $\lambda_j = 1/n_j$*

$$\sum_{j=0}^{M} \lambda_j \, f(u_j) = \int \varrho(u) \, f(u) \, \mathrm{d}u \qquad \textit{for every polynomial } f \textit{ of degree at most } 2M+1\textit{:} \tag{27}$$

the u_j and λ_j are the nodes and weights of the Gauss rule of the spectral density $\varrho \, \mathrm{d}u$, and $a_j = \sqrt{\lambda_j}\, \mathcal{T}[G](u_j)$ for states in the span of the retained moments.

Appendix C gives the proofs. A truncated hierarchy is therefore exactly $M+1$ passive slices at the Gauss nodes of the Case–Van Kampen measure, and the results above hold for it with sums over j in place of integrals over u. At $\Lambda = \infty$ the nodes are the Gauss–Hermite nodes (Parker and Dellar, 2015; Tassi, 2022); in KRMHD the Landau coupling enters only A_{10} of a matrix that does not depend on the wavenumber, and the Alfvénic fields do not depend on g, so the decoupling is exact in three dimensions. The largest phase velocity $u_{\max} = u_M$ hardly depends on Λ; with $q_m = (2m+1)^{1/2}$,

$$\begin{aligned} u_{\max} &= q + 2^{-1/3}\alpha_1 \, q^{-1/3} - \frac{2^{-2/3}\alpha_1^2}{10} \, q^{-5/3} + O(q^{-3}), && q = q_{M+1} \quad \text{(zero closure)}, \\ u_{\max} &= q + 2^{-1/3}\alpha_1' \, q^{-1/3} - 2^{-5/3}\Big(\frac{4}{5\alpha_1'} + \frac{\alpha_1'^2}{5}\Big) q^{-5/3} + O(q^{-3}), && q = q_M \quad \text{(copy closure)}, \end{aligned} \tag{28}$$

where $\alpha_1 = -2.3381074$ and $\alpha_1' = -1.0187930$ are the first zeros of the Airy function and of its derivative (Szegő, 1975; Olver, 1959, theorem 6.32); $u_{\max} = 7.25$ at $M = 32$. Slice j is advected with the stream function $\Phi + s u_j \Psi$, so an explicit time step for the nonlinear term must resolve the speed $\max|\nabla_\perp \Phi| + s u_{\max} \max|\nabla_\perp \Psi|$, not $\max|\nabla_\perp \Phi| + s \max|\nabla_\perp \Psi|$; the simulations of section 7 set their time step in this way.

5 Frozen field lines

When the field lines are frozen into the $\boldsymbol{E} \times \boldsymbol{B}$ flow, which is common to all slices (11), each slice streams, in the frame of the flow, along field lines that do not move. This Lagrangian linearity is due to S09 (§6.3), who showed with the Cauchy solution for the perturbed field that in the frame of $\boldsymbol{u}_E$ the derivative $\hat{\boldsymbol{b}} \cdot \nabla$ is taken along the field lines at $t = 0$ (their eqs 214–216), and concluded that the compressive fluctuations do not cascade along the moving field lines; Kunz et al. (2015, §§2.6 and 4.1) restated it for pressure-anisotropic plasmas, and Lithwick and Goldreich (2001, §2.3.1) made the corresponding remark for a passive scalar in MHD turbulence. We give the solution operator and the hypotheses, and use volume preservation to obtain exact statements about volume-integrated statistics such as $W(m)$.

5.1 The frozen-line theorem

We treat one branch at a time, with state $g(\boldsymbol{r}, t)$: the function $G(\boldsymbol{r}, \cdot, t)$ of u, or $(g_0, \ldots, g_M)$ in a truncated hierarchy, where A replaces L. With collisions $-\mathcal{C}g$ as in (18), (7) reads

$$\partial_t g + \{\Phi, g\} + s \, \nabla_\parallel (Lg) = -\mathcal{C}g. \tag{29}$$

Let $D_t = \partial_t + \{\Phi, \cdot\}$ and $\nabla_\parallel(t) = \partial_z + \{\Psi(\cdot, t), \cdot\}$, let $\boldsymbol{X}_t$ be the flow map of $\boldsymbol{u}_E$, with $\partial_t \boldsymbol{X}_t(\boldsymbol{\xi}) = \boldsymbol{u}_E(\boldsymbol{X}_t(\boldsymbol{\xi}), t)$ and $\boldsymbol{X}_0(\boldsymbol{\xi}) = \boldsymbol{\xi}$, and let $\mathcal{X}_t h = h \circ \boldsymbol{X}_t^{-1}$ turn a function of the labels $\boldsymbol{\xi}$ into a function

of position. The first of equations (2) states that the residual $R = \partial_t\Psi + \{\Phi,\Psi\} - \partial_z\Phi$ vanishes. The hypotheses are

(H1) *Ideal induction.* On $[0,t_1]$ the residual R depends on z and t only. The vorticity equation may be forced, viscous or prescribed.

(H2) *Kinetic equation.* The branch has $\Lambda \notin [0,1]$, and g obeys (29), where $\mathcal{C}$ is a linear operator on the velocity variable, the same at every point and independent of the fields, and is bounded or dissipative. There is no spatial dissipation of g and no source.

(H3) *Smoothness.* Φ, Ψ and all their spatial derivatives are continuous on $\mathbb{T}^3\times[0,t_1]$, except that Φ may jump at finitely many times.

(30)

Under (H3), $\boldsymbol{X}_t$ is a volume-preserving diffeomorphism of $\mathbb{T}^3$. The frozen generator $L_0 = s\,\nabla_\parallel(0)\,L + \mathcal{C}$ describes the linear problem with the field lines fixed in their initial shape and no flow, and e^{-tL_0} is its solution operator.

Theorem 3 (Frozen-line theorem)**.** *Assume (30), and let g be a classical solution of (29) on $[0,t_1]$. Then*

$$g(t) = \mathcal{X}_t\, e^{-tL_0}\, g(0), \qquad 0 \le t \le t_1. \tag{31}$$

Proof. With $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}}\times\nabla_\perp\Psi$, $\partial_t\hat{\boldsymbol{b}} - \nabla\times(\boldsymbol{u}_E\times\hat{\boldsymbol{b}}) = \hat{\boldsymbol{z}}\times\nabla R$, which vanishes under (H1), so $\hat{\boldsymbol{b}}$ obeys the ideal induction equation and, since $\nabla\cdot\boldsymbol{u}_E = \nabla\cdot\hat{\boldsymbol{b}} = 0$, has Cauchy's solution (Webb et al., 2014, example 3) $\hat{\boldsymbol{b}}(\boldsymbol{X}_t(\boldsymbol{\xi}),t) = (\partial\boldsymbol{X}_t/\partial\boldsymbol{\xi})\,\hat{\boldsymbol{b}}(\boldsymbol{\xi},0)$. The chain rule then gives, for a differentiable function h of the labels,

$$\nabla_\parallel(t)\,\mathcal{X}_t h = \mathcal{X}_t\,\nabla_\parallel(0) h; \tag{32}$$

equivalently, by the Jacobi identity,

$$[D_t, \nabla_\parallel]\, f = \{R, f\}, \tag{33}$$

which vanishes under (H1). Write $g(t) = \mathcal{X}_t\hat{g}(t)$. By (29), $\partial_t\hat{g} = -[s\nabla_\parallel(t)(Lg) + \mathcal{C}g]\circ\boldsymbol{X}_t$. The operators L and $\mathcal{C}$ act at each point with kernels independent of $\boldsymbol{r}$ (for L because the field relations of KRMHD are algebraic), so they commute with composition by $\boldsymbol{X}_t$, and (32) gives $\partial_t\hat{g} = -L_0\hat{g}$ with $\hat{g}(0) = g(0)$. For $\Lambda\notin[0,1]$, $s\nabla_\parallel(0)L$ is antisymmetric in the free-energy norm, which gives uniqueness. □

Without collisions, (31) decouples into slices,

$$a_u(t) = \mathcal{X}_t\, T_{sut}\, a_u(0), \qquad T_\ell = e^{-\ell\nabla_\parallel(0)}, \tag{34}$$

where T_ℓ moves a function a distance ℓ along the field lines of $\hat{\boldsymbol{b}}(0)$: each slice is carried rigidly along the initial field lines at the speed su and then relabelled by the flow. This makes exact, for the slices, the remark of S09 (§6.2.4) that the ballistic response is also the nonlinear solution in Lagrangian variables. Whether solutions of the ideal three-dimensional RMHD equations stay smooth is not known in general, so (H3) is an assumption on $[0,t_1]$.

5.2 Single-point statistics and forcing

Let O be a function of the local state, all the velocity-space information at one point, and of finitely many of its derivatives along the field. We call

$$\mathcal{O}_t[f] = \int O(f, \nabla_\parallel(t)f, \nabla_\parallel(t)^2 f, \dots)\,\mathrm{d}^3r \tag{35}$$

a single-point statistic. Volume preservation and $\nabla_\parallel(t)^k\mathcal{X}_t = \mathcal{X}_t\nabla_\parallel(0)^k$ give $\mathcal{O}_t[\mathcal{X}_t y] = \mathcal{O}_0[y]$.

Corollary 1 (Single-point statistics)**.** *Assume (30), and let* $h(t) = e^{-tL_0} g(0)$*. Then* $\mathcal{O}_t[g(t)] = \mathcal{O}_0[h(t)]$ *for every single-point statistic (35) and* $0 \le t \le t_1$*: every volume-integrated single-point statistic of* g *takes the value it has in the linear problem along the field lines frozen at* $t = 0$*, with no flow.*

The corollary covers $W(m,t)$, the slice correlations $C(u,u';t)$, the collisional dissipation rate and the Hermite flux (20) with the full $\nabla_\parallel$. It does not cover spectra in $k_\perp$ or k_z, fluxes resolved in $k_\perp$ shells or computed with ∂_z, which involve Fourier filters that do not commute with $\mathcal{X}_t$. With a source F, Duhamel's formula in the labels shows that the source injected at t' phase-mixes along the field lines frozen at t' and is then relabelled by the flow. For forcing that is white in time this gives the following.

Corollary 2 (White forcing)**.** *Let the forcing be* $F\,\mathrm{d}t = \mathrm{d}Y$*, with* Y *a Wiener process with spatially smooth realisations and covariance operator* Σ *whose increments are independent of the past of* g *and of the Alfvénic fields, let (H1) and (H3) hold for every realisation, and let* g *otherwise satisfy (H2). For every single-point statistic whose integrand is a quadratic form,*

$$\mathrm{E}\,\mathcal{O}_t[g(t)] = \mathrm{E}\,\mathcal{O}_0\big[e^{-tL_0} g(0)\big] + \int_0^t \mathrm{E}\,\mathcal{O}_{t'}\big[e^{-(t-t')L_{t'}}\chi\big]\,\mathrm{d}t', \tag{36}$$

where $L_{t'} = s\,\nabla_\parallel(t')\,L + \mathcal{C}$ *and* χ *is a random field with zero mean and covariance* Σ*, independent of all other variables. The mean of* $W(m,t)$ *is therefore the sum of the linear responses to the initial condition and to the individual injections, each phase-mixing along the field lines frozen at its injection time.*

Sketch of proof. In the labels, $\mathrm{d}\hat{g} = -L_0\hat{g}\,\mathrm{d}t + \mathcal{X}_t^{-1}\mathrm{d}Y$, whose solution is an Itô integral with an integrand fixed by the past. The cross term between the initial condition and the stochastic integral has zero mean, the Itô isometry turns the rest into $\int_0^t \mathrm{E}\,\mathcal{O}_0[e^{-(t-t')L_0}\mathcal{X}_{t'}^{-1}\chi]\,\mathrm{d}t'$, and relabelling with (32) gives (36). □

The corollary concerns the mean: a single realisation contains cross terms between injections at different times, and for forcing correlated in time the mean cross terms between injections closer than the correlation time survive.

5.3 Field-aligned wavenumbers

Since $\hat{\boldsymbol{b}}(t)$ is smooth and divergence-free, $K_\parallel(t) = -\mathrm{i}\,\nabla_\parallel(t)$ is self-adjoint on $L^2(\mathbb{T}^3)$ (e.g. Kato, 1966). Its spectral variable is the field-aligned wavenumber, and in a uniform field its spectral projections are the Fourier filters in k_z.

Corollary 3 (Field-aligned wavenumbers)**.** *Assume (30), and let* $E_t(I)$ *be the spectral projection of* $K_\parallel(t)$ *on a Borel set* I*. Then*

$$E_t(I)\,g(t) = \mathcal{X}_t\,e^{-tL_0}\,E_0(I)\,g(0). \tag{37}$$

For real g *the parts with positive and negative field-aligned wavenumber evolve separately, so no part of* g *changes the sign of its field-aligned wavenumber, and without collisions the field-aligned spectrum of every slice is constant in time.*

Proof. By (32) and the unitarity of $\mathcal{X}_t$, $E_t(I) = \mathcal{X}_t E_0(I)\mathcal{X}_t^{-1}$, and $E_0(I)$ commutes with e^{-tL_0}. Without collisions e^{-tL_0} multiplies the component of the slice at u at field-aligned wavenumber $k_\parallel$ by $e^{-\mathrm{i}suk_\parallel t}$. □

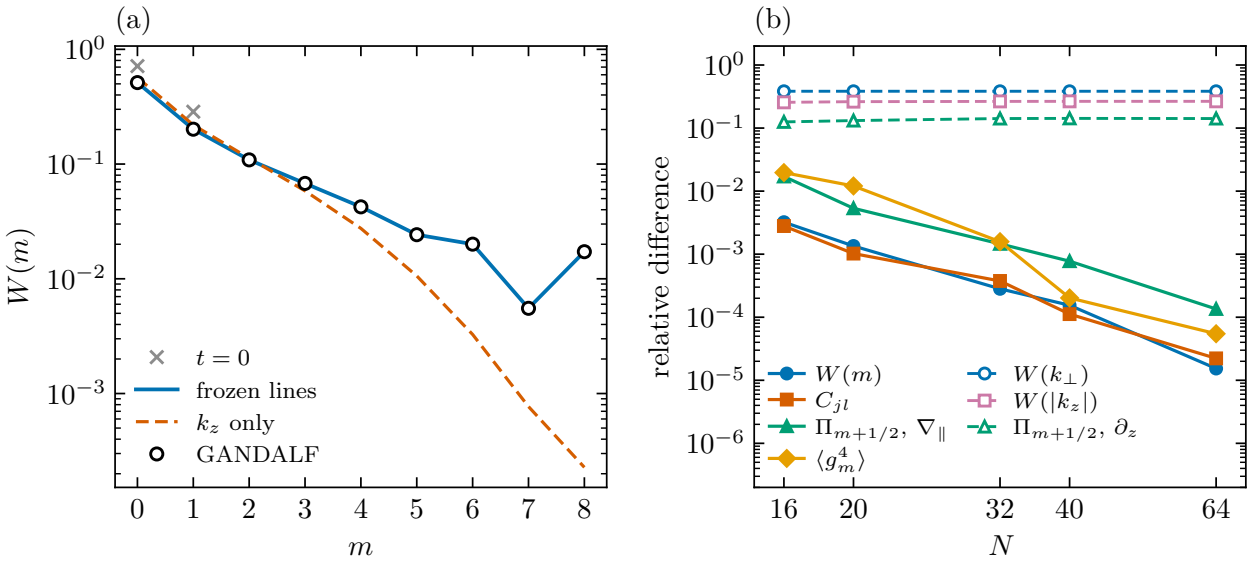


Figure 2: Test of the frozen-line theorem. Ideal GANDALF runs (no dissipation, forcing or collisions) from one initial state at N^3 grid points, with $M = 8$ (zero closure), $\Lambda = \sqrt{5}$, $\beta_i = 1$, random g_0 and g_1, $\chi = 2.21$, the term $\{\Phi, g_0\}$ 2.91 times the streaming term in rms, and $\Delta t = 0.0005$. (a) The Hermite spectrum $W(m)$ at $t = 0.125$ for $N = 64$: initial state (crosses), GANDALF (circles), frozen-line prediction (solid line) and k_z-only prediction (dashed line). (b) The relative difference between GANDALF and the frozen-line prediction at $t = 0.125$ against N, for the single-point statistics $W(m)$, C_{jl}, $\Pi_{m+1/2}$ with $\nabla_\parallel$ and $\langle g_m^4 \rangle$ (filled markers), and for $W(k_\perp)$, $W(|k_z|)$ and $\Pi_{m+1/2}$ with ∂_z (open markers).

The Eulerian k_z of the same part of g can change sign. In the frozen-in shear $\Phi = -Szy$, $\Psi = -Sty$, an exact but not periodic solution of (2) with $\nabla_\parallel(t) = \partial_z + St\,\partial_x$, a Fourier mode of the labels becomes a Fourier mode of position with

$$k_z(t) = k_z - Stk_x, \qquad k_\parallel(t) = k_z(t) + Stk_x = k_z. \tag{38}$$

Its k_z changes sign at $t = k_z/(Sk_x)$ while its field-aligned wavenumber stays fixed, so a diagnostic that sorts the free energy by the sign of k_z counts the mode as reversed, although nothing has changed along the field.

5.4 What breaks the theorem

Resistivity or hyper-resistivity on Ψ, parallel dissipation of $z^\pm$, and forcing of Ψ in general make R depend on x and y, so that (33) does not vanish and the field lines are not carried by the flow. Dissipation of g that depends on $k_\perp$ or k_z does not commute with the relabelling and violates (H2). Fourier truncation does not commute with it either; its error vanishes as the resolution increases at fixed t, but not uniformly in t. In the equations that GANDALF integrates, the hyper-resistivity acts alike on $z^\pm$ and every g_m and violates (H1) and (H2), the hypercollisions satisfy (H2), and forcing that adds the same increment to z^+ and z^- keeps (H1).

5.5 A numerical check

The runs of figure 2 integrate one ideal initial state at $N = 16$ to 64 for about three eddy times and compare each run with the frozen-line prediction $e^{-tL_0}g(0)$, computed with GANDALF's right-hand side for g with $\Phi = 0$ and $\Psi = \Psi(0)$ held fixed, and with the k_z-only prediction ($\Phi = \Psi = 0$). For the statistics covered by corollary 1 the relative difference falls by two to three orders of magnitude from $N = 16$ to 64, to between 1.5×10^{-5} for $W(m)$ and 1.4×10^{-4} for the field-aligned flux, while for $W(k_\perp)$, $W(|k_z|)$ and the flux with ∂_z it stays between 0.13 and 0.38 at every N. The k_z-only prediction misses $W(m)$ by 15% and the field-aligned flux by 37%. In ideal turbulence forced through Φ, the response of $W(m)$ to each white injection into g converges with N to phase mixing along the field lines frozen at its injection time, as corollary 2 states. With one term added, the departure from the frozen-line prediction still falls with N for viscosity or forcing of Φ, but not for resistivity or forcing of Ψ or for damping of g in $k_\perp$ or k_z.

5.6 Echoes

Under (30), $W(m, t)$ and the field-aligned Hermite flux evolve as in the linear problem $\partial_t h = -L_0 h$, so a return of free energy to low m must already be present there, as a linear echo of initial unmixing correlations, the recurrence of a truncated hierarchy, or slow phase mixing at small field-aligned wavenumbers; with white forcing the same holds for $\mathrm{E}\, W(m, t)$. In this sense ideal, smooth dynamics without spatial dissipation of g produces no stochastic echoes in $W(m)$. In drift kinetics with a straight guide field (Schekochihin et al., 2016; Parker et al., 2016) the particles stream along a fixed direction, $[D_t, \partial_z]f = -\{\partial_z\Phi, f\} \neq 0$, and the z-dependent flow couples positive and negative $k_\parallel$, which is the mechanism of the stochastic echo; a flow independent of z gives none (Schekochihin et al., 2016, appendix B; Kanekar, 2015, §5.2). In KRMHD the shear $\partial_z \boldsymbol{u}_E$ that would change $k_\parallel$ also bends the field lines together with g, so under (H1) the commutator (33) vanishes and no part of g moves between positive and negative field-aligned wavenumbers (corollary 3), although its Eulerian k_z can change sign (38).

The theorem does not contradict the simulations in which echoes were reported (Meyrand et al., 2019), which, like ours, dissipate both the magnetic field and the compressive distribution in space and so violate (H1) and (H2). An echo in $W(m)$ requires such terms, or a failure of deterministic flux freezing for rough fields as the resistivity goes to zero, which (H3) excludes; Eyink et al. (2013) reported such a failure at high conductivity, and Eyink (2018, §VIII.C) argued that in a turbulent inertial range the ideal Ohm's law does not imply flux freezing. The question for turbulence is therefore what the terms that break (30) do to $W(m)$. The next section proposes an answer.

6 A two-channel model of the Hermite spectrum

In a turbulent steady state the compressive distribution is forced and dissipated, and the hypotheses (30) fail in two ways: the induction equation is not ideal, and g is dissipated in space. The exact results still constrain what the nonlinearity can do. It moves no free energy between slices (section 4.1), and as long as the field lines stay frozen into the flow it moves none between field-aligned wavenumbers (corollary 3). This section builds a model on the assumption that the two failures matter only through the removal of free energy at small perpendicular scales, and lists the predictions by which it can be tested.

6.1 The two channels

Phase mixing at a frozen field-aligned wavenumber. With white forcing, the mean of $W(m)$ is a sum of linear responses to the individual injections, each phase mixing along the field lines frozen at its injection time (corollary 2). Each injection has, along those lines, a distribution of field-aligned wavenumbers κ, which it keeps (corollary 3). By (24), free energy injected at κ moves up the Hermite ladder at the speed $\mathrm{d}m/\mathrm{d}t = s\kappa\sqrt{2m}$ and carries the flux

$$\Pi_{m+1/2} \simeq s\kappa\sqrt{2m}\, W(m) \qquad (m \gg 1), \tag{39}$$

with no part returning to lower m.

A uniform sink. Each slice is a passive scalar, which the turbulence carries to the dissipation scale, where the spatial dissipation removes it, slice by slice (section 4.1). For passive scalars in turbulence the rate at which variance is dissipated stays finite as the diffusivity goes to zero (Shraiman and Siggia, 2000; Falkovich et al., 2001), and the time a scalar takes to reach the dissipation scale is set by the flow, not by the diffusivity. All slices share the $\boldsymbol{E}\times\boldsymbol{B}$ flow, and in its frame they differ only by their streaming along the field lines (theorem 3), which moves them along the lines but does not by itself refine them across the lines. We therefore assume that the turbulence cascades every slice in the same way. If the perpendicular spectrum of g_m then has the same shape for every m, the dissipation, which acts alike on every moment, removes free energy from each moment in proportion to its free energy,

$$D_\perp(m) = \frac{W(m)}{T}, \tag{40}$$

with one time T for all m. Equivalently, for single-point statistics such as $W(m)$, the nonlinearity of KRMHD acts as a damping at a rate $1/T$ that does not depend on m.

In the steady state of (21), treated as continuous in m, (39) and (40) give, without collisions,

$$\frac{\mathrm{d}}{\mathrm{d}m}\bigl[s\kappa\sqrt{2m}\, W(m)\bigr] = -\frac{W(m)}{T} \qquad\Longrightarrow\qquad W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}}\exp\Bigl(-\frac{\sqrt{2m}}{s\kappa T}\Bigr), \tag{41}$$

where ε is the rate of injection into g_0. The free energy at m was injected a time $\sqrt{2m}/(s\kappa)$ earlier and has survived the sink for that long. Because the injected pattern is fixed in the Eulerian frame and the field lines are tilted, the injected free energy has a distribution $P(\kappa)$ of field-aligned wavenumbers, and

$$W(m) = \varepsilon\int P(\kappa)\,\frac{1}{s\kappa\sqrt{2m}}\exp\Bigl(-\frac{\sqrt{2m}}{s\kappa T}\Bigr)\,\mathrm{d}\kappa, \tag{42}$$

with each κ breaking at its own m, so that a broad $P(\kappa)$ turns the stretched exponential into a more gradual decline. For a single κ the local slope is $\mathrm{d}\ln W/\mathrm{d}\ln m = -\frac{1}{2}(1+\sqrt{m/m_*})$, which passes through -1 at $m = m_*$ and $-3/2$ at $m = 4m_*$, so the spectrum is close to m^{-1} around the break without being a power law. A spread of κ makes the decline more gradual, because the free energy at large κ phase mixes fastest and survives to the largest m; for a distribution with a Gaussian tail of width σ, (42) falls at very large m as $\exp[-c\,(m/s^2T^2\sigma^2)^{1/3}]$ with c of order one, more slowly than the single-κ exponential. With the distribution of the injected pattern measured at $\beta_i = 1$ (mean $1.22\,k_1$, rms $1.39\,k_1$), the local slope of (42) stays within 0.15 of -1 over a factor of five in m around m_* and reaches only -2 at $16\,m_*$ and -2.5 at $64\,m_*$, against -2.5 and -4.5 for a single κ. A spectrum measured over a limited range near m_* can therefore look like a power law somewhat steeper than m^{-1}. Hypercollisions add the sink $D_\nu(m)$ to (41). For the comparison with simulations we solve, for each κ, the forced linear hierarchy (9) with $\nabla_\parallel$ replaced by $\mathrm{i}\kappa$, with

the damping rate $1/(2T)$ added to every amplitude and with the hypercollisions of the run, and sum over the measured $P(\kappa)$ (appendix D). This reproduces (42) at large m and is exact for the linear problem at small m. Once T and $P(\kappa)$ are measured, the model has no free parameter.

The form (41) is that of Zocco and Schekochihin (2011, eq. 112) for the Hermite spectrum of growing modes, with their 2γ replaced by $1/T$, and Schekochihin et al. (2016, eqs 2.58–2.59) solved the same continuity equation in m with the collisional sink $2\nu m$ in place of $1/T$. What the model adds is the identification of the sink with the passive-scalar cascade, which the exact results of sections 4 and 5 make the only effect of the nonlinearity on $W(m)$ once the field lines are frozen.

6.2 Scales

The spectrum (41) follows $m^{-1/2}$ up to the break

$$m_* = \frac{(s\kappa T)^2}{2} = \frac{C^2}{2}\,\frac{\beta_i}{\chi^2}, \tag{43}$$

where the second form takes $T = C/(k_\perp u_\perp)$ and $\kappa \simeq k_\parallel$ at the outer scale, $s = \beta_i^{1/2}$ with $v_A = 1$, and the nonlinearity parameter $\chi = k_\perp u_\perp/(k_\parallel v_A)$. This is the phase-mixing threshold $k_\parallel v_{\rm th} \sim \sqrt{m}\,k_\perp u_\perp$ of Schekochihin et al. (2016, eq. 4.5), written in KRMHD variables, with its coefficient tied to the cascade time; for $\chi \sim 1$ it is the crossover $m \sim \beta_i$ stated by Kawazura et al. (2019). Beyond m_* the model gives an exponential decay in $\sqrt{m}$, not a power law. If the collisional cutoff m_c lies below m_*, most of the injected free energy reaches it and is dissipated by collisions; if it lies above, most is dissipated at small perpendicular scales, and the collisional share is about $\exp(-\sqrt{2m_c}/(s\kappa T))$.

By (25), the exponential factor in (41) is the cosine transform at $q = \sqrt{2m}$ of a Lorentzian correlation between slices with width

$$\delta u_c = \frac{1}{s\kappa T} = \frac{1}{\sqrt{2m_*}}. \tag{44}$$

Two slices loaded with the same pattern dephase at the rate $s\kappa|u - u'|$ and are removed at the rate $1/T$, so in the steady state slices closer than δu_c in phase velocity remain correlated.

6.3 Predictions

The model predicts the following, each measurable independently of the others.

(i) *Uniform sink.* $D_\perp(m)/W(m)$ is independent of m, and the budget (21) closes with the flux (20) computed with the full $\nabla_\parallel$, but not with the k_z-only flux.

(ii) *Shape.* With T from (i) and $P(\kappa)$ measured along field lines for the injected pattern, $W(m)$ and the field-aligned flux follow the model. A return flux would make both fall below it at large m.

(iii) *Dependence on β_i.* At fixed χ the Alfvénic turbulence does not depend on β_i, so T and $P(\kappa)$ are unchanged and $m_* \propto \beta_i$: the break moves into a fixed range of m as β_i decreases.

(iv) *Anomaly.* T stays finite and does not depend on the dissipation coefficients as the resolution increases.

(v) *Eulerian diagnostics.* A split of the flux by the sign of k_z assigns to the backward direction the forward flux of modes whose k_z and field-aligned wavenumber have opposite signs (38),

and so reports a backward flux where the field-aligned flux has none. If the free energy at moment m phase mixes at field-aligned wavenumber κ while the split sorts it by $k_z = \kappa - \delta$, with $\delta = \boldsymbol{b}_\perp \cdot \boldsymbol{k}_\perp$ Gaussian with variance $\sigma^2 = \langle k_z^2 \rangle - \kappa^2$, the expected ratio of the unmixing to the phase-mixing part is $[\sigma\phi(z) - \kappa(1 - \Phi(z))]/[\kappa\Phi(z) + \sigma\phi(z)]$ with $z = \kappa/\sigma$ and ϕ, Φ the standard normal density and distribution; it grows with the tilt and so with the range of resolved $k_\perp$.

(vi) *Slice correlations.* The correlation between slices falls off over δu_c (44).

The echo picture (Meyrand et al., 2019) predicts instead a return flux along the field, which reduces the net field-aligned flux and $W(m)$ below the model at large m, and a field-aligned wavenumber of g that grows with $k_\perp$ through the inertial range as g inherits the parallel cascade of the Alfvénic fields.

6.4 Limits of the model

The model rests on two assumptions that the exact results do not guarantee. First, it assumes that the field lines stay frozen long enough for the injected κ to be kept, although the hyper-resistivity of any simulation, and resistivity in a plasma, reconnect them. Meyrand et al. (2019) argued that this relabelling gives g the parallel structure of the Alfvénic fields within about one eddy-turnover time. Second, it assumes that the perpendicular cascade has a dissipative anomaly. Drivas and Eyink (2017) showed that anomalous dissipation of a passive scalar requires Lagrangian spontaneous stochasticity, the non-uniqueness of trajectories as the diffusivity goes to zero, and Eyink (2018, §VIII.C) argued that the same stochasticity breaks flux freezing in a turbulent inertial range. If the two go together, the frozen κ of the first assumption must fail at the scales where the anomaly of the second operates. The model can survive this only if the free energy that phase mixes sits at scales where the field lines are still effectively frozen and the relabelling acts on free energy that the perpendicular sink is already removing. Section 7 measures the field-aligned wavenumber of g as a function of $k_\perp$ to see where this holds. The model also takes the cascade rate to be the same for every slice; slices with large $|u|$ stream fast along wandering field lines, which may refine them across the lines faster than slow slices, and prediction (i) measures how large this effect is. At 128^3 the field-aligned wavenumber of g_0 grows through the inertial range, roughly as $k_\perp^{1/3}$, more slowly than that of the Alfvénic fields, and the sink rate falls by 30% from 64^3 (section 7.6), so both assumptions hold only approximately at the resolutions reached here.

If the sink depends on the time a since injection rather than being uniform, as section 7.5 finds, free energy injected at κ reaches m at the age $a = \sqrt{2m}/(s\kappa)$, and (41) becomes

$$W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}} \exp\Big(- \int_0^{\sqrt{2m}/(s\kappa)} \gamma(a)\, \mathrm{d}a \Big), \tag{45}$$

with $\gamma(a)$ the rate at which the perpendicular cascade removes free energy of age a. For a passive scalar injected at the outer scale, γ is small at first, while the cascade carries the fresh pattern towards the dissipation scale, and approaches a constant once the scalar's spectrum has filled in. A uniform $1/T$ then removes too much free energy at small m and too little at large m. The function $\gamma(a)$ belongs to the Alfvénic turbulence, not to β_i, so at fixed χ one $\gamma(a)$ should describe every β_i.

7 Simulations

7.1 Set-up

We integrate (2) in Elsasser form with one compressive branch (9), the branch Λ^+ of (6) with $\tau = Z = 1$, truncated at $M = 32$ with the zero closure, using GANDALF (version 0.8.1; Kanekar, 2025) in double precision. The runs at $M = 32$ used version 0.7.0, whose solver is the same, on a CPU; the runs at $M = 64$ (section 7.6) used a GPU. The box is periodic with side $L = 1$ and 64^3 grid points, so $k_1 = 2\pi$; $v_A = 1$ and $\tau_A = L/v_A = 1$. This section reports the run at $\beta_i = 1$, where $\Lambda^+ = \sqrt{5}$; section 7.5 describes the runs at lower β_i.

The Alfvénic fluctuations are forced through Φ alone, so that z^+ and z^- receive the same increment, which keeps the forcing out of the induction equation (section 5). The forcing is white in time and acts on the Fourier modes with $1 \le k_\perp/k_1 \le 2$ and $|k_z| = k_1$, with injection rate $\varepsilon_A = 0.3$. Excluding $k_z = 0$ from the band avoids the two-dimensional condensate that a band including it drove at this resolution. The density moment g_0 is forced on the same band, white in time, with its amplitude rescaled at every step so that the expected injection rate is $\varepsilon = 0.3$; the band excludes $k_\perp = 0$, where injected free energy would phase mix with no perpendicular cascade. Every injection therefore has the single Eulerian parallel wavenumber $k_z = \pm k_1$, and its field-aligned wavenumbers differ from k_1 only through the tilt of the field lines.

GANDALF applies the hyper-resistivity $\eta\,(k_\perp^2/k_{\perp,\max}^2)^2$, with $\eta = 300$, alike to $z^\pm$ and to every g_m; there is no parallel dissipation. Hypercollisions $\nu\,(m/M)^6$ with $\nu = 30$ act on g_m for $m \ge 2$. Free energy moving up the ladder at $\mathrm{d}m/\mathrm{d}t = s\kappa\sqrt{2m}$ is attenuated by them mainly above $m = M/2$, so the range $m \le M/2$ is nearly collision-free. The time step is GANDALF's Lawson–Runge–Kutta scheme (Lawson-RK4; Lawson, 1967), which integrates the streaming exactly through an integrating factor and treats the nonlinear terms with explicit RK4. Following section 4.4, $\Delta t = 0.8 \times 2\sqrt{2}/\rho$, where ρ is the spectral radius of the explicit slice-advection operator at the extreme nodes $u = \pm u_{\max}$, computed by a Lanczos iteration every 25 steps, or the Alfvénic limit $0.8\,\Delta x/\max|\nabla_\perp z^\pm|$ if that is smaller. The Alfvénic turbulence is first spun up with the same forcing and dissipation. The run then starts from $g = 0$ and lasts $53\,\tau_A$, and all averages are over the final $50\,\tau_A$, $3 \le t \le 53$; W reaches its mean level within about $3\,\tau_A$.

We characterise the outer scale by the part of Φ in the shells $k_\perp/k_1 \le 2.5$. Its energy-weighted mean $k_\perp$ and $|k_z|$ are $k_{\perp,\mathrm{out}} = 9.7 = 1.55\,k_1$ and $k_{\parallel,\mathrm{out}}$, and its rms $\boldsymbol{E}\times\boldsymbol{B}$ velocity is $u_{\mathrm{out}} = 0.71$. The nonlinearity parameter is $\chi = k_{\perp,\mathrm{out}}u_{\mathrm{out}}/k_{\parallel,\mathrm{out}} = 0.99$ (0.85 with the rms field-aligned wavenumber of the same field in place of $|k_z|$), and the outer eddy time is $\tau_{\mathrm{nl}} = 1/(k_{\perp,\mathrm{out}}u_{\mathrm{out}}) = 0.144$. The measured free-energy injection rate over the window is $\varepsilon = 0.307$.

7.2 Diagnostics

The energy change of every step is split into the parts due to the ideal dynamics, the forcing, the hyper-resistivity and the hypercollisions. This gives $D_\perp(m)$ and $D_\nu(m)$ in (21) exactly for every moment, and the remainder is the time-stepping error of the ideal part. Every 20 steps we record $W(m)$, its perpendicular spectrum $W(m, k_\perp)$, the field-aligned flux (20) and the k_z-only flux, the split of the k_z-only flux into phase-mixing and unmixing parts by the sign of k_z (Schekochihin et al., 2016), the slice amplitudes a_j at the $M+1$ nodes of theorem 2 with their correlation matrix $C_{jl} = \langle a_j a_l\rangle$, and the Alfvénic energy $E_A = \frac{1}{2}\langle|\nabla_\perp\Phi|^2 + |\nabla_\perp\Psi|^2\rangle$ and its spectrum.

Field-aligned wavenumbers are measured along traced field lines, as is done for the Alfvénic fields (Cho and Vishniac, 2000; Meyrand et al., 2019). On nine snapshots ($t = 10, 15, \ldots, 50$) we trace 4096 lines of $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}} \times \nabla_\perp\Psi$ over four box lengths in z, sample g_m, Φ and Ψ along them,

Table 1: Checks of the $\beta_i = 1$ run over the window $3 \le t \le 53$.

Check	Alfvénic	Compressive
Time-stepping error / injected energy	5.0×10^{-4}	3.8×10^{-8}
|injection − dissipation| / injection	1.7%	0.06%
Standard deviation / mean of E_A and W	0.117	0.141
Spread of block means, four blocks of $12.5\,\tau_A$	2.4% (E_A)	8.8% ($W(m)$), 6.9% (Π), 15.5% ($D_\perp$), 8.4% (D_ν)
Energy fraction beyond $k_\perp/k_1 = 21$	4×10^{-5}	2×10^{-5}
Free energy at $m > 3M/4$	–	1.1%
Share of D_ν at $m \ge M/2$	–	95%

the last three also filtered into the bands $k_\perp/k_1 \in [0,2]$, $[2,4]$, $[4,8]$ and $[8,16]$, and compute the spectra along the lines. We report κ_1, the first moment of $|\kappa|$. The distribution $P(\kappa)$ of the injected free energy is measured in the same way, on fresh realisations of the density forcing sampled along the same lines. Appendix D gives the details and the computation of the model.

7.3 Validation

Table 1 lists the checks. Energy is conserved by the time step and balanced between injection and dissipation. The Alfvénic energy fluctuates by 12% about its mean, which at this box size is intrinsic to the forced turbulence. The block means of the averages used below, each divided by the injection rate of its block, agree to within 9% except $D_\perp$ at $m = 0$ (15.5%), so the averages are converged to about that level. The perpendicular spectrum $E_A(k_\perp)$ follows $k_\perp^{-5/3}$ only over $2 \le k_\perp/k_1 \le 8$ and is steeper than $k_\perp^{-4}$ from $k_\perp/k_1 = 9$, so the inertial range is short. The hierarchy is marginally resolved: 1.1% of the free energy lies at $m > 3M/4$. We therefore treat this run as a pilot, which tests predictions (i), (ii), (v) and (vi) but cannot place the break m_* inside the hierarchy (section 7.4).

7.4 Results at $\beta_i = 1$

(i) A uniform sink. Figure 4(b) shows $D_\perp(m)/W(m)$. It lies between 0.37 and 0.47 for $0 \le m \le 16$, varies by a factor of 1.20 between its extremes and by 6% in rms over $1 \le m \le 16$, and rises to 0.54 at $m = 24$, where the hypercollisions dominate. We take $1/T = 0.437$, its median over $2 \le m \le 16$, so $T = 2.29 = 15.9\,\tau_{\rm nl}$. The budget (21) closes to $4 \times 10^{-4}\,\varepsilon$ in every moment with the field-aligned flux, and to 8% with the k_z-only flux. The perpendicular dissipation is therefore close to an m-independent sink, as (40) assumes, with a systematic rise of 20% over $1 \le m \le 16$. Two causes are possible: slices with large $|u|$, which dominate the high moments, may be cascaded faster (section 6.4), or the free energy at higher m, which was injected longer ago, has had more time to reach small perpendicular scales. The runs at lower β_i (section 7.5) decide between them: at a fixed time since injection the streaming speed of the slices that dominate moment m, $s\sqrt{2m}$, scales as s^2 and differs by a factor of ten between $\beta_i = 1$ and 0.1, yet the sinks of all runs fall on one curve against that time (figure 7b). The rise is an effect of age.

Field-aligned wavenumbers. The injected pattern, which has $k_z = k_1$, has $\kappa_1 = 1.14\,k_1$ along the field lines; the rms of the tilt $\boldsymbol{b}_\perp \cdot \boldsymbol{k}_\perp$ over the pattern is $0.89\,k_1$, and the rms field-aligned wavenumber is $1.34\,k_1$. The moments g_m have κ_1 between $1.07\,k_1$ and $1.19\,k_1$ for $1 \le m \le 21$, with no trend in m, and the wavenumber implied by the flux, $\Pi_{m+1/2}/[s\sqrt{2(m+1)}\,W(m)]$, lies between $1.02\,k_1$ and $1.10\,k_1$ for $3 \le m \le 23$. Figure 5 resolves κ_1 in $k_\perp$. In the bands centred on $k_\perp/k_1 = 1.5$,

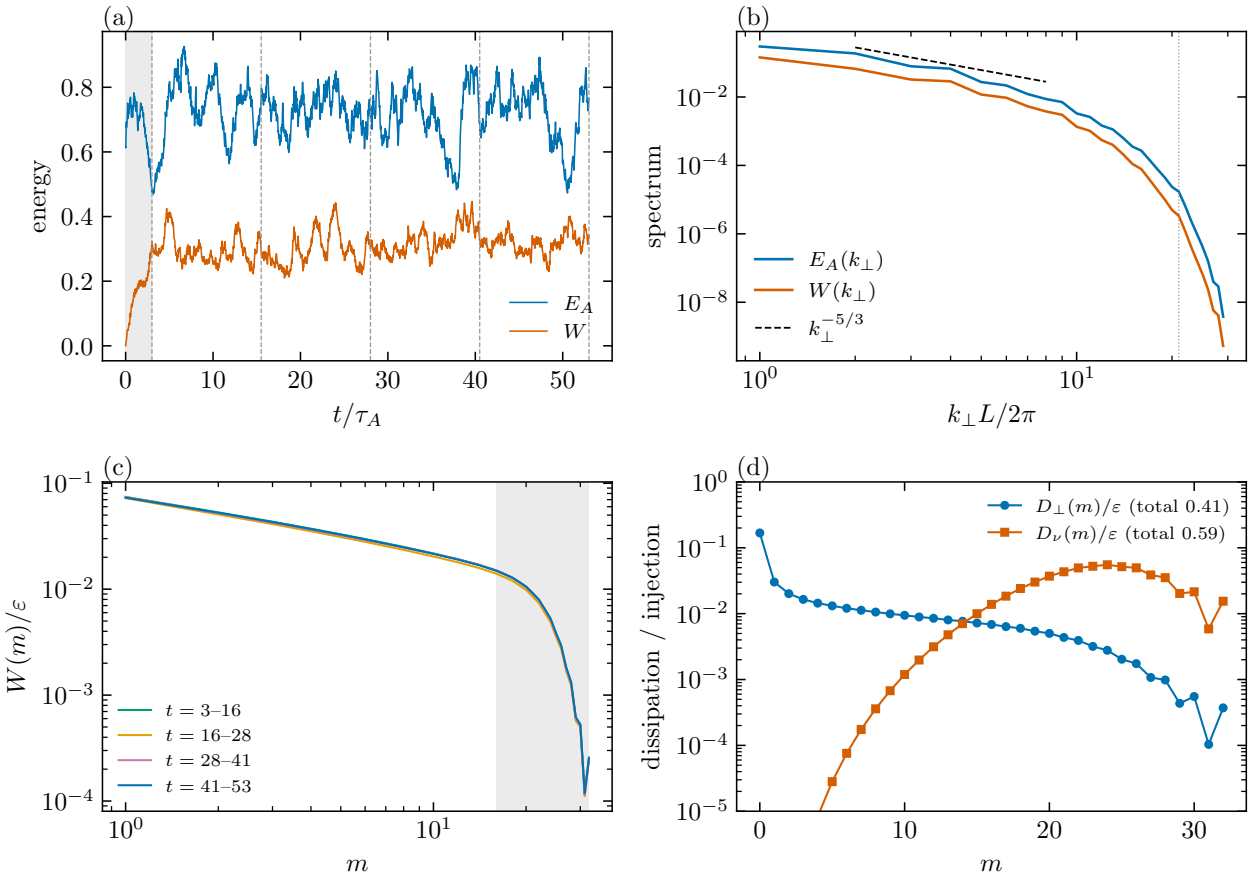


Figure 3: Overview of the run at $\beta_i = 1$, 64^3, $M = 32$. (a) E_A and W against t; the dashed lines bound the four averaging blocks, and the shading marks the excluded start. (b) $E_A(k_\perp)$ and $W(k_\perp)$ averaged over $3 \le t \le 53$, with $k_\perp^{-5/3}$ for reference; the dotted line is the edge of the dealiasing band. (c) $W(m)/\varepsilon$ in each block, divided by the injection rate of that block; the shading marks $m > M/2$, where the hypercollisions act. (d) The rates at which the hyper-resistivity, $D_\perp(m)$, and the hypercollisions, $D_\nu(m)$, remove free energy from moment m, as fractions of the injection rate; in total they remove 41% and 59% of it.

3, 6 and 12, g_0 has $\kappa_1/k_1 = 0.91$, 0.97, 1.36 and 2.04, whereas Φ has 1.11, 1.55, 2.12 and 3.18 and Ψ has 1.13, 1.47, 1.94 and 2.60. The Alfvénic fields gain parallel structure from the outer scale on, though more slowly than $k_\perp^{2/3}$ at this resolution. The density moment keeps the injected field-aligned wavenumber up to $k_\perp = 3k_1$, exceeds it by 20% in the band $4 \le k_\perp/k_1 \le 8$ at the end of the inertial range, and reaches $2.04\,k_1$ only in the band $8 \le k_\perp/k_1 \le 16$, where E_A is steeper than $k_\perp^{-4}$. Measured with k_z, g_0 would appear to have $\kappa_1/k_1 = 0.78$, 0.93, 1.83 and 3.12, close to the Alfvénic values at small scales. In this run, then, g acquires substantial structure along the field only at the end of the inertial range and in the dissipation range, where the hyper-resistivity reconnects the field lines and the perpendicular sink is already removing it. At 128^3, with a longer inertial range, g_0 gains field-aligned structure more gradually and throughout it (section 7.6).

(ii) The shape of $W(m)$ and the flux. With T from (i) and $P(\kappa)$ of the injected pattern, the model reproduces $W(m)$ to within -7% and $+8\%$ for $1 \le m \le 30$, with an rms deviation of 3.3% in $\ln W$ over $2 \le m \le 16$ (figure 4a, green), and the field-aligned flux to within 3–7% for $m \le 24$ (figure 6a). Neither falls below the model at large m; the measured flux lies slightly above it. The

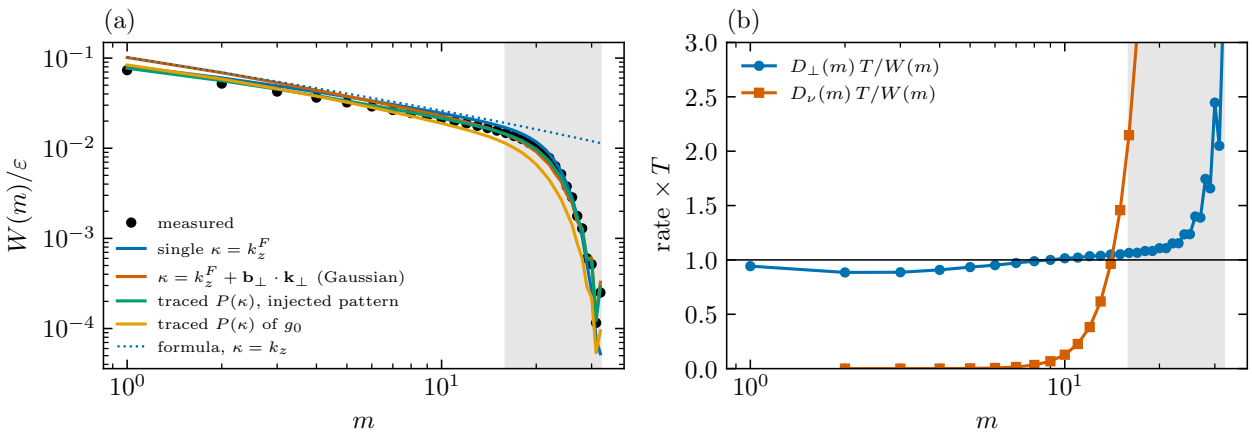


Figure 4: (a) The Hermite spectrum $W(m)/\varepsilon$ at $\beta_i = 1$ (black dots) and the two-channel model with the measured T and four choices of κ: a single $\kappa = k_z^F = k_1$, the wavenumber of the forcing (blue); $\kappa = k_z^F + \boldsymbol{b}_\perp \cdot \boldsymbol{k}_\perp$ with a Gaussian distribution of the tilt (red); the distribution $P(\kappa)$ of the injected pattern traced along field lines (green); and the distribution of g_0 itself traced along field lines (yellow). The dotted line is the formula (41) with $\kappa = k_z$ and no collisions. (b) The rates at which the perpendicular dissipation (circles) and the hypercollisions (squares) remove free energy from moment m, divided by $W(m)$ and multiplied by T. The shading marks $m > M/2$, where the hypercollisions act.

flux that $W(m)$ would carry if all of it phase mixed at its measured $\kappa_1(m)$ is close to the measured flux, so there is no room for a return flux of more than a few per cent. With a single $\kappa = k_1$ the model gives W 14–16% too high, because it omits the tilt. With the field-aligned distribution of g_0 itself in place of that of the injected pattern, it overdamps, giving W too low by a factor of up to 1.9 at $m = 24$: the free energy phase mixes at the wavenumber at which it was injected, not at the larger one that g_0 shows at small perpendicular scales.

The break. With these values the break lies far beyond the hierarchy: $m_* = (s\kappa T)^2/2$ is 103 for $\kappa = k_1$ and 185 for the rms field-aligned wavenumber, and $C^2\beta_i/(2\chi^2) = 127$ with $C = T/\tau_{\rm nl}$. Free energy injected at $m = 0$ reaches $m = 16$ after about $0.8\,\tau_A$ and $m = 32$ after $1.1\,\tau_A$, against $T = 2.3\,\tau_A$, and the hypercollisions remove 57–61% of the injected free energy (the range over the four blocks). At $\beta_i = 1$ the exponential factor in (41) therefore reduces W by only 29% at $m = 16$ and 39% at $m = 32$, and the test of the shape is correspondingly weak: it confirms that there is no deficit, not that the sink produces a break.

(v) Eulerian diagnostics. The k_z-only flux split by the sign of k_z has an unmixing part of -0.11 to -0.14 times its phase-mixing part for $m \geq 1$, and -0.52 at $m = 0$ (figure 6a), whereas the field-aligned flux is reproduced by a model that has no return flux. A diagnostic of this kind would report a backward flux of about 12% of the forward one where, by the field-aligned measures, there is none. With the rms field-aligned and Eulerian wavenumbers of g_m measured in the run, about $1.2\,k_1$ and 1.8–$1.9\,k_1$, the estimate of prediction (v) gives a ratio of 0.09–0.11 for $m \geq 1$, against the measured 0.11–0.15: the backward flux of the split is accounted for by the tilt of the field lines, with no field-aligned return flux (figure 8).

(vi) Slice correlations. The width (44) is $\delta u_c = 0.05$–0.07, below the spacing of the nodes, about 0.4 near $u = 0$ at $M = 32$, so it cannot be resolved here. The model, evaluated at the nodes, reproduces the measured normalised correlations $C_{jl}/(C_{jj}C_{ll})^{1/2}$ to an rms difference of 0.027 (figure 6b).

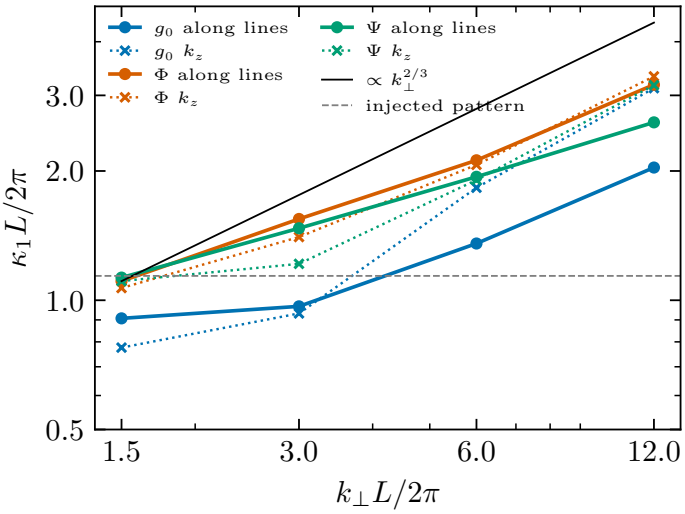


Figure 5: The mean field-aligned wavenumber κ_1 of g_0 (blue), Φ (red) and Ψ (green) in four bands of $k_\perp$, measured along traced field lines (solid) and from the Eulerian k_z (dotted), at $\beta_i = 1$. The dashed line is the injected pattern traced along the same lines; the thin line is proportional to $k_\perp^{2/3}$.

7.5 Dependence on β_i at fixed χ

To bring the break into the hierarchy, we ran the same set-up at $\beta_i = 0.3$ and 0.1, each at $M = 32$ and $M = 64$, and at $\beta_i = 1$ also at $M = 64$. The Alfvénic sector does not depend on β_i, so the same Alfvénic forcing gives the same χ (the outer-scale value is 0.95–0.99 in every run); the branch is $\Lambda^+ = 6.2206$ and 19.198. The attenuation of free energy that streams up the ladder by hypercollisions $\nu(m/M)^6$ depends on $\nu\sqrt{M}/s$, so we set $\nu = 30\,\beta_i^{1/2}$ at $M = 32$ and $\nu = 30\,\beta_i^{1/2}(32/M)^{1/2}$ at $M = 64$, which keeps the total attenuation fixed while the collisional range moves from $m > 16$ to $m > 32$. With T and $P(\kappa)$ from $\beta_i = 1$, (43) gives $m_* \approx 31$–56 at $\beta_i = 0.3$ and $m_* \approx 10$–19 at $\beta_i = 0.1$. The runs at $\beta_i = 0.1$ and at $M = 64$ use a safety factor of 0.4 on the Alfvénic limit of the time step, with both limits updated every step, and last $66\,\tau_A$, with averages over $16 \le t \le 66$.

Table 2 lists the runs. All conserve energy to within 2.1×10^{-3} of the injection and balance injection and dissipation to within 2%. The free energy fluctuates more slowly as β_i decreases: its integrated correlation time is about 0.8, 1.4 and $3.6\,\tau_A$ at $\beta_i = 1$, 0.3 and 0.1 (at $M = 32$), so a window of $50\,\tau_A$ holds fewer independent samples, and at $\beta_i = 0.1$ the four block means differ by up to 23%, about the scatter expected from that correlation time. We quote the block spread as the uncertainty of the averages, about ±12% on $W(m)$ at $\beta_i = 0.1$. In the run at $\beta_i = 0.3$ and $M = 32$ the free energy was still settling during the first $12\,\tau_A$ of the nominal window, which we therefore start at $t = 15.5$. The spectra show no sign of numerical instability at small scales in any run.

The break enters the hierarchy. At $\beta_i = 0.1$ the predicted break, $m_* \approx 6$–10 with the measured T, lies well inside the resolved range, and $W(m)$ falls below $m^{-1/2}$ from small m (figure 7a). The continuum formula (41) with $\kappa = k_1$ and the measured T follows the measured spectrum to about 10% up to $m \approx 16$, where the hypercollisions take over at $M = 32$; at $M = 64$ the spectrum continues to fall steeply between $m \approx 15$ and 40. At $\beta_i = 0.3$ the spectrum bends at correspondingly larger m. To measure the break we fit $W(m)\sqrt{2m} \propto \exp(-\sqrt{2m}/\lambda)$ over $2 \le m \le M/2$, below the collisional range. The fitted λ is 16.8, 8.3 and 3.8 at $\beta_i = 1$, 0.3 and 0.1 ($M = 32$), and 15.7, 7.2

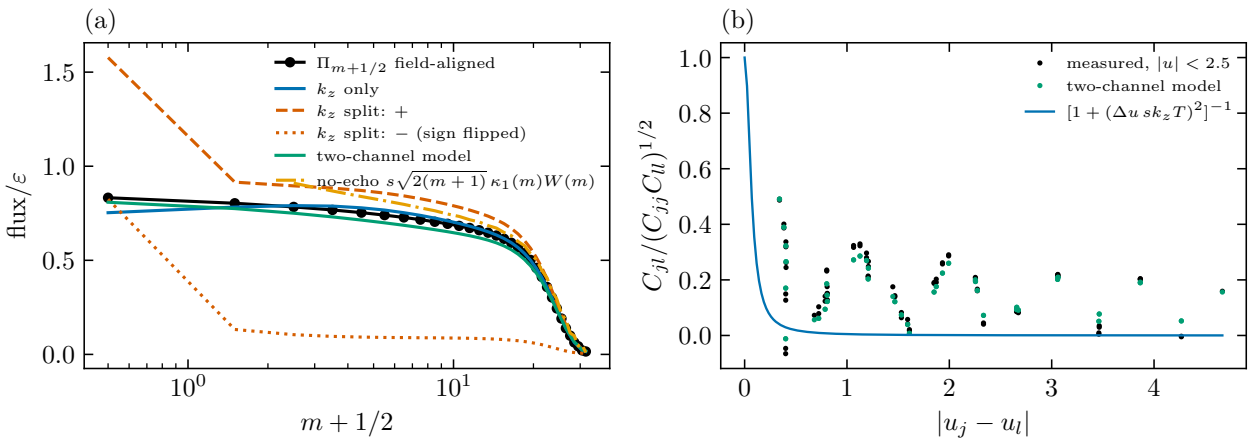


Figure 6: (a) Hermite fluxes divided by ε at $\beta_i = 1$: the field-aligned flux (20) (black), the k_z-only flux (blue), its phase-mixing part (dashed) and its unmixing part with the sign reversed (dotted), the two-channel model (green), and the flux $s\sqrt{2(m+1)}\,\kappa_1(m)\,W(m)$ that $W(m)$ would carry if all of it phase mixed at the measured field-aligned wavenumber (dash-dotted). (b) Correlations $C_{jl}/(C_{jj}C_{ll})^{1/2}$ between slices with $|u| < 2.5$ against $|u_j - u_l|$: measured (black) and two-channel model (green); the line is the Lorentzian $[1 + (\Delta u\, sk_zT)^2]^{-1}$ of width (44).

and 3.6 at $M = 64$, against $s\kappa_1 T = 16.4$, 8.0, 3.8, 16.8, 7.6 and 3.7 from the measured T and the injected κ_1: the measured decay rate in $\sqrt{2m}$ is the predicted one to within 7% in every run. The break $m_* = \lambda^2/2$ falls from about 140 at $\beta_i = 1$ to about 7 at $\beta_i = 0.1$, by a factor of about 20 for a factor of 10 in β_i, because $m_* \propto \beta_i T^2$ and T itself decreases at low β_i (below).

No return flux, and a sink that depends on age. At every β_i the measured $W(m)$ lies at or above the model with the traced $P(\kappa)$, never below it, so there is no sign of a return flux. At $\beta_i = 1$ the model fits to within 8%. At lower β_i the measured spectrum exceeds it, by up to 21% at $\beta_i = 0.3$ and 39% at $\beta_i = 0.1$ at $M = 32$, and by up to 14%, 31% and 50% at $\beta_i = 1$, 0.3 and 0.1 at $M = 64$, where T is taken over the larger collision-free range $2 \le m \le 32$. The excess comes from the assumption of a uniform sink: $D_\perp(m)/W(m)$ rises with m, from 0.33–0.43 at $m \le 1$ in every run to 0.46, 0.53 and 0.62 at $m = 16$ for $\beta_i = 1$, 0.3 and 0.1, so a single rate $1/T$, the median over $2 \le m \le 16$, removes too much free energy at small m. The rise is steeper at lower β_i, where free energy reaches a given m later. Plotted against the time $a = \sqrt{2m}/(s\kappa_1)$ since injection, the sinks of all runs fall on one curve (figure 7b), with an rms scatter of 4.5% about a joint fit $D_\perp/W = 0.50 + 0.12 \ln a$, against 12.6% when plotted against m. The sink is about 0.4 for the youngest free energy and levels off at about 0.63 for ages beyond about $2.5\,\tau_A$. The perpendicular sink is therefore a property of the age of the free energy, the same at every β_i at fixed χ, as expected for a passive scalar whose variance reaches the dissipation scale only after the cascade has had time to act (section 6.4). The median rate $1/T$ rises from 0.44 to 0.59 between $\beta_i = 1$ and 0.1 for the same reason: at low β_i the moments $2 \le m \le 16$ hold older free energy. [*To do:* the two-channel model with the measured age-dependent sink $\gamma(a)$ in place of $1/T$, fitted at one β_i and used to predict the others]

Diagnostics. The k_z-only flux split by the sign of k_z has an unmixing part of -0.11 to -0.15 times its phase-mixing part for $m \ge 1$ at every β_i, which the tilt of the field lines accounts for to within 0.01–0.05 for $m \ge 4$ with no return flux (figure 8), while the per-m budget closes to better than $5 \times 10^{-4}\,\varepsilon$ with the field-aligned flux in every run, and to 2–8% with the k_z-only flux. The slice correlation width, fitted at the nodes, is 0.30 at $\beta_i = 0.3$ and 0.38 at $\beta_i = 0.1$, against the

Table 2: The forced runs, on N^3 grid points with M Hermite moments. The window is the averaging interval in units of τ_A; std/mean is that of E_A and W over the window; the block spread is the largest relative spread of the four block means of $W(m)$, Π, $D_\perp(m)$ and $D_\nu(m)$; $W_>$ is the fraction of the free energy at $m > 3M/4$; $1/T$ is the median of $D_\perp(m)/W(m)$ over $2 \le m \le 16$, the range below the collisional cutoff in every run.

β_i	N	M	window	std/mean (E_A, W)	block spread	$W_>$	$1/T$	$T/\tau_{\rm nl}$
1	64	32	3–53	0.117, 0.141	0.155	1.1%	0.437	15.9
0.3	64	32	15.5–53	0.076, 0.147	0.174	1.0%	0.478	13.5
0.1	64	32	16–66	0.120, 0.180	0.229	0.65%	0.590	11.2
1	64	64	16–66	0.146, 0.143	0.312[a]	1.2%	0.421	15.4
0.3	64	64	16–66	0.158, 0.136	0.185	0.87%	0.508	13.2
0.1	64	64	16–66	0.114, 0.186	0.212	0.44%	0.604	11.0
0.3	128	64	16–66	0.114, 0.156	0.236[a]	1.06%	0.352	18.4
1	128	64	16–66	[*To do:*]	[*To do:*]	[*To do:*]	[*To do:*]	[*To do:*]

[a] $D_\perp(m)$ at $m = 0$ (31% and 24%; 20% at $m = 1$ for $\beta_i = 1$); at most 19% for every other average and m.

predicted 0.11–0.14 and 0.22–0.30; the node spacing near $u = 0$ is about 0.4 at $M = 32$, so it is not resolved. At $M = 64$, where the spacing is about 0.28, the fitted widths are 0.25 and 0.35 against the predicted 0.12–0.16 and 0.24–0.32, so at $\beta_i = 0.1$ the width approaches the prediction as it becomes resolved.

7.6 Dependence on resolution

Doubling M leaves the spectra in the range both runs resolve nearly unchanged (figure 9). For $1 \le m \le 10$, $W(m)$ at $M = 64$ is within 1.01–1.05 of its value at $M = 32$ at $\beta_i = 1$, 0.96–0.99 at $\beta_i = 0.3$ and 1.00–1.01 at $\beta_i = 0.1$, inside the block uncertainty. Between $m = 12$ and 16 the runs at $M = 32$ fall below those at $M = 64$, by up to 7% beyond that offset at $\beta_i = 1$ and 5% at $\beta_i = 0.3$ and 0.1, because there the hypercollisions of the $M = 32$ runs, at the rate $\nu(m/M)^6$, begin to compete with the perpendicular sink; at $m = 16$ and $\beta_i = 1$ the two rates are equal. The rate $1/T$ changes by -4%, 6% and 2% at $\beta_i = 1$, 0.3 and 0.1. What changes is the share of the free energy that reaches the hypercollisions: it falls from 59% to 50% at $\beta_i = 1$, from 43% to 27% at $\beta_i = 0.3$ and from 20% to 10% at $\beta_i = 0.1$ when the collisional range moves from $m > 16$ to $m > 32$. The model predicts this. Free energy injected at κ reaches the collisional cutoff m_c with probability $\exp(-\sqrt{2m_c}/(s\kappa T))$, which with $m_c = M/2$, the measured T and $\kappa = \kappa_1$ gives 71% and 63% at $\beta_i = 1$, 50% and 36% at $\beta_i = 0.3$, and 23% and 12% at $\beta_i = 0.1$. The model reproduces the factor by which the share falls and overestimates its level by 15–30%, as the age-dependent sink, stronger at the large ages that m_c corresponds to, would make it. The collisional share of the dissipation is therefore not a converged quantity at $M = 32$: at fixed β_i and χ, free energy that would be dissipated by collisions at a low cutoff is removed at small perpendicular scales when the cutoff is raised.

Doubling the perpendicular resolution at $\beta_i = 0.3$ and $M = 64$, with the hyper-resistivity normalised to the grid, lengthens the inertial range: the spectral slope of E_A over the fitted shells is -3.5 at 128^3 against -4.1 at 64^3. The outer scale is unchanged ($\tau_{\rm nl} = 0.150$ and 0.154, $\chi = 0.97$ and 0.93). The perpendicular sink is not (figure 10): $D_\perp(m)/W(m)$ is 25–30% lower at every m, $1/T$ falls from 0.51 to 0.35 and $T/\tau_{\rm nl}$ rises from 13.1 to 18.4, and the share of the free energy dissipated by hypercollisions rises from 27% to 38%. The spectrum still decays as the model says,

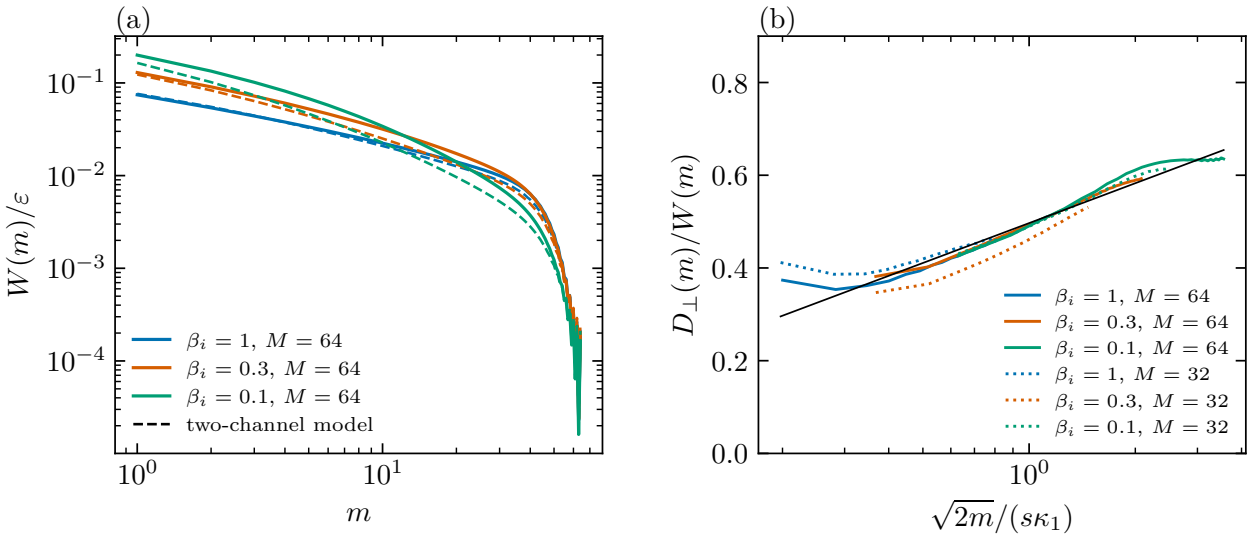


Figure 7: (a) $W(m)/\varepsilon$ at $\beta_i = 1$, 0.3 and 0.1 ($M = 64$) (solid) and the two-channel model with the traced $P(\kappa)$ of the injected pattern and the measured T of each run (dashed). (b) The perpendicular sink $D_\perp(m)/W(m)$ against the time $\sqrt{2m}/(s\kappa_1)$ since injection of the free energy at m, with κ_1 the mean field-aligned wavenumber of the injected pattern, for $1 \le m \le M/2$; the runs at $M = 64$ (solid) reach older free energy than those at $M = 32$ (dotted); the thin line is the joint fit $0.50 + 0.12 \ln[\sqrt{2m}/(s\kappa_1)]$.

with a fitted $\lambda = 9.7$ against $s\kappa_1 T = 10.9$, so the break moves out with T, from $m_* \approx 26$ to about 47. $W(m)$ and the field-aligned flux still lie at or above the model, by up to 27% and 24%, and the per-m budget closes to $6 \times 10^{-5}\,\varepsilon$.

The field-aligned wavenumber of g_0 also changes. In the bands centred on $k_\perp/k_1 = 1.5$, 3, 6 and 12 it is 1.02, 1.27, 1.52 and $2.07\,k_1$ at 128^3, against 0.98, 1.08, 1.37 and $2.04\,k_1$ at 64^3, while Φ has 1.08, 1.66, 2.18 and $3.12\,k_1$. With the longer inertial range, g_0 gains structure along the field gradually across it, roughly as $k_\perp^{1/3}$ against about $k_\perp^{1/2}$ for Φ, rather than only at the dissipation scale. Measured with k_z, its wavenumber in the band at $12\,k_1$ would be $3.68\,k_1$, so Eulerian measures still overstate the parallel structure of g. The unmixing part of the k_z-split flux grows from about -0.13 to between -0.20 and -0.25 times the phase-mixing part. The rms Eulerian wavenumber of g_m grows from about $1.9\,k_1$ to $2.7\,k_1$ while its field-aligned wavenumber stays at about $1.27\,k_1$, and the tilt estimate of prediction (v) gives 0.23–0.29 (figure 8): the growth of the backward flux of the split is the growth of the tilt at the resolved scales, while the field-aligned flux shows no deficit.

At the resolutions we reach, then, neither the rate of the perpendicular sink nor the field-aligned wavenumber of g is independent of the dissipation scale, and prediction (iv) is not confirmed. Two readings remain. At 64^3 the dissipation range reached almost to the forcing scales and made the sink artificially fast, in which case T should level off at higher resolution; or the perpendicular cascade of the slices has no dissipative anomaly, in which case T, and with it m_*, keep growing as the dissipation weakens. A third resolution would decide between them. [*To do:* the 128^3 run at $\beta_i = 1$]

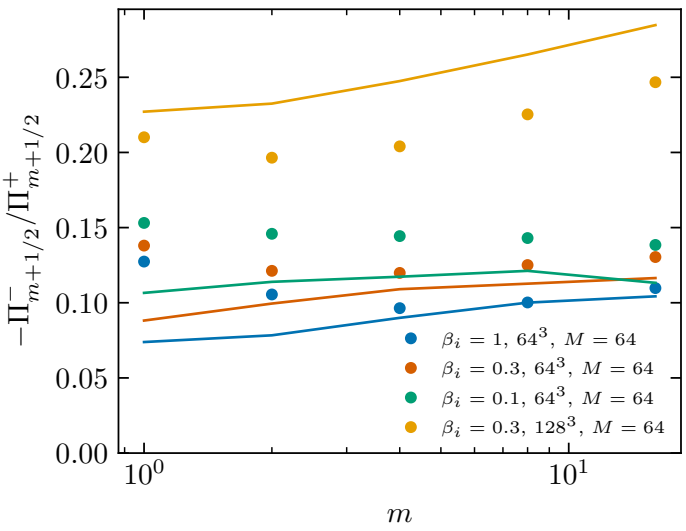


Figure 8: The unmixing part of the k_z-only Hermite flux split by the sign of k_z, relative to its phase-mixing part, measured (markers) and estimated from the tilt of the field lines with no return flux (lines; prediction (v)), for the runs at $M = 64$.

8 Discussion

Two results of this paper are exact. The compressive sector of KRMHD is a continuum of passive slices labelled by phase velocity, for each branch and in truncated hierarchies, for any smooth Alfvénic fields and without collisions (theorems 1 and 2), so the nonlinearity moves no free energy between phase velocities. With an ideal induction equation, smooth fields and no spatial dissipation of the compressive distribution, every volume-integrated single-point statistic, including $W(m)$ and the field-aligned Hermite flux, evolves as in linear phase mixing along the initial field lines (theorem 3), so such dynamics produces no stochastic echoes in $W(m)$.

The two-channel model extends these results to turbulence, where they do not apply directly, by treating spatial dissipation as a sink that removes free energy from every Hermite moment at the passive-scalar cascade rate. In the pilot simulation at $\beta_i = 1$ and $\chi \approx 1$ the model holds to the accuracy of the measurement. The perpendicular dissipation is close to an m-independent sink, and $W(m)$ and the field-aligned flux are those of phase mixing at the injected field-aligned wavenumber with that sink, with no deficit at large m. The density moment acquires substantial structure along the field only at the perpendicular scales where the hyper-resistivity acts. In this run the relabelling of field lines that Meyrand et al. (2019) invoke happens, but only to free energy that is being removed at the same scales, and the free energy that phase mixes keeps its injected wavenumber.

Our spectrum at $\beta_i = 1$ differs from that of Meyrand et al. (2019), who found $W(m)$ somewhat steeper than m^{-1} at the same β_i and a net flux to high m much smaller than the injection rate. [*To do:* compare the two set-ups: forcing (fields, band, correlation time), dissipation of g ($k_\perp^4$ here, $k_\perp^8$ there), resolution in $k_\perp$ and m, and the flux diagnostic (summed over $k_\perp$ here, per $k_\perp$ shell there); state which differences could change $W(m)$ and which runs would decide it] Two points are already clear. A diagnostic built on the Eulerian k_z reports a backward flux of about 12% of the forward flux in our run, where the field-aligned measures show none, so splits and fluxes of the free energy should be computed along the perturbed field. And the field-aligned wavenumber of g measured with k_z grows with $k_\perp$ much as that of the Alfvénic fields does, whereas measured along the field

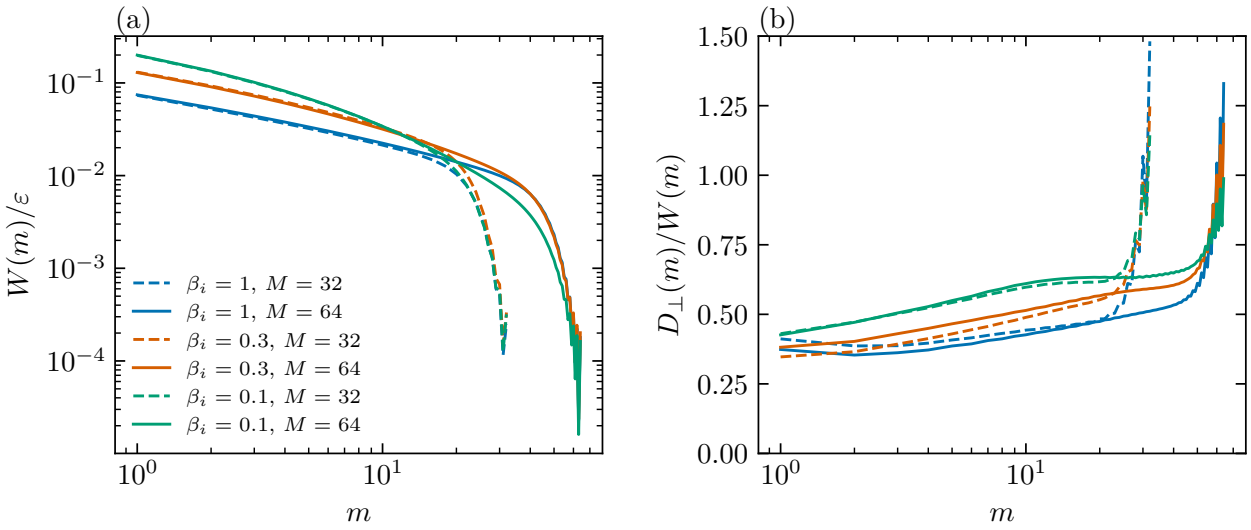


Figure 9: Resolution in Hermite index at $\beta_i = 1$ (blue), 0.3 (orange) and 0.1 (green): $M = 32$ (dashed) and $M = 64$ (solid), with the same box, forcing, seeds and hyper-resistivity and the hypercollision coefficient scaled as described in section 7.5. (a) $W(m)/\varepsilon$. (b) $D_\perp(m)/W(m)$.

lines it stays close to the injected value through most of the inertial range.

The model predicts that m_* grows in proportion to β_i at fixed χ. At $\beta_i = 1$ the break lies beyond the hierarchy, at $\beta_i = 0.1$ it lies inside it and $W(m)$ steepens from small m, as predicted, with the same Alfvénic turbulence in every run. The trend that Kawazura et al. (2019, 2020) found in gyrokinetic simulations, $m^{-1/2}$ at high β_i and close to m^{-1} at $\beta_i = 0.1$, is the trend the model gives without any echo, because m_* falls with β_i. Fitted to the measured spectra, the decay rate in $\sqrt{2m}$ is the predicted $1/(s\kappa_1 T)$ to within 7% at every β_i and both M, and the break moves from $m_* \approx 140$ at $\beta_i = 1$ to $m_* \approx 7$ at $\beta_i = 0.1$. The comparison also showed where the model is too simple: the perpendicular sink is not uniform in m but grows with the age of the free energy, and with that refinement (45) the sinks of all runs collapse onto one function of age. The model then also predicts, and the runs at $M = 32$ and $M = 64$ confirm, that the share of the free energy dissipated by collisions falls as the collisional cutoff m_c rises, as $\exp(-\sqrt{2m_c}/(s\kappa T))$. If this holds as $m_c \to \infty$, the free energy that phase mixes is in the end removed at small perpendicular scales, and the fraction dissipated by collisions in velocity space at fixed β_i and χ is set by where the collisions act; in a real plasma the perpendicular cascade of the slices stops at the ion gyroscale, beyond the reach of KRMHD.

The main open question is the limit of high resolution. The model needs a perpendicular cascade whose rate does not depend on the dissipation coefficients and field lines that stay frozen for the free energy that phase mixes. Drivas and Eyink (2017) and Eyink (2018) give reasons to expect that the first requires the failure of the second at small enough scales (section 6.4). Between 64^3 and 128^3 at $\beta_i = 0.3$, T rose from 13 to 18 outer eddy times and g_0 gained field-aligned structure gradually through the longer inertial range (section 7.6). Neither the dissipative anomaly nor the frozen wavenumber is therefore established at these resolutions. The model's description of $W(m)$ in terms of the measured T and κ still holds, and the absence of a return flux is unchanged, but the asymptotic values of T and m_* are open, and a third resolution is needed to tell a sink that converges from one that keeps weakening. [*To do:* the 128^3 run at $\beta_i = 1$]

The results hold for $k_\perp \rho_i \to 0$. At finite $k_\perp \rho_i$ the Landau coupling depends on the wavenum-

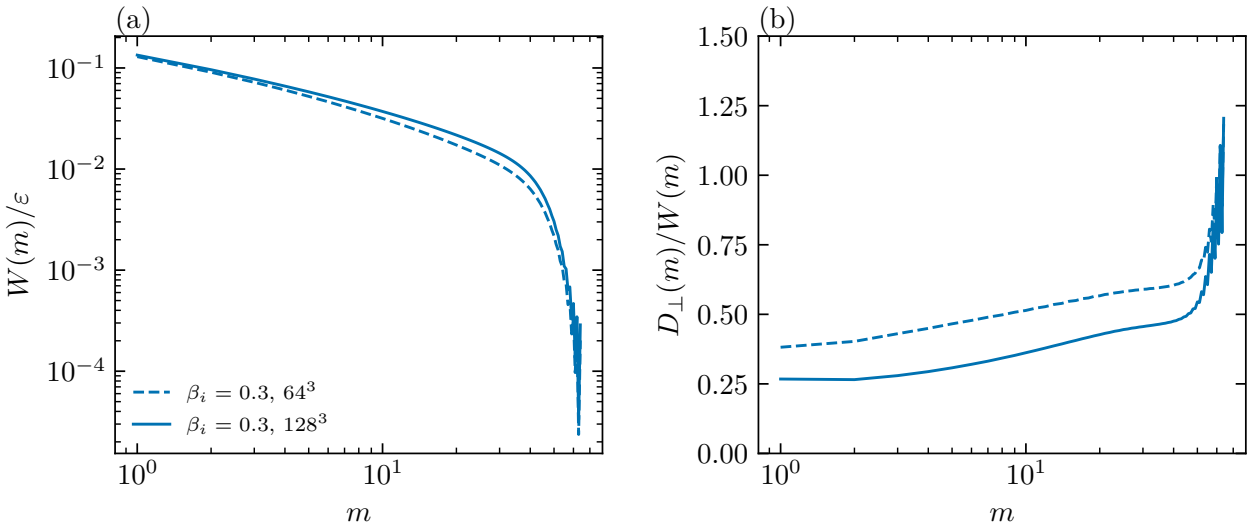


Figure 10: Resolution in $k_\perp$ at $\beta_i = 0.3$ and $M = 64$: 64^3 (dashed) and 128^3 (solid), with the hyper-resistivity normalised to the grid. (a) $W(m)/\varepsilon$. (b) $D_\perp(m)/W(m)$. [*To do:* add $\beta_i = 1$]

ber, the advection depends on $v_\perp$ and the compressive fluctuations act back on the Alfvénic ones (Chandran et al., 2026), so the slices no longer decouple. Velocity-dependent collisions are the only terms in KRMHD that exchange free energy between slices (18).

A The two compressive branches

We derive the branches of section 2.2 following S09 (§6.2) and CMM (appendix B), with $\delta_n \equiv \delta n/n$, $\delta_B \equiv \delta B/B$ and $\langle\cdot\rangle$ the volume average. We checked every identity symbolically, and the moments and splits also by quadrature over $(v_\perp, v_\parallel)$ for random distribution functions.

Table 3: Conventions of this paper, CMM (Chandran et al., 2026), S09 (Schekochihin et al., 2009) and GANDALF (Kanekar, 2025), with equation numbers of the respective papers; "same" means as in this paper.

This paper	CMM	S09	GANDALF
$v_{\rm th} = (2T_i/m_i)^{1/2}$, $s = v_{\rm th}/v_A = \beta_i^{1/2}$	same	same	$s = \beta_i^{1/2}$
$F_M(u) = \pi^{-1/2} e^{-u^2}$, $u = v_\parallel/v_{\rm th}$	same, in $v_\parallel$ (below B4)	as CMM	same
$z^\pm = \Phi \pm \Psi$; z^+ propagates toward $-z$	same by (3.23)–(3.24)	same (21)	same
$\Psi = -A_\parallel/(4\pi n_i m_i)^{1/2}$	same (3.2)	same (135)	$\Psi = (z^+ - z^-)/2$
$\Lambda = \Lambda^\pm$ of (6); $\Lambda = \infty$ for $l \geq 2$	same (B7)	same (182)	input
$g_m = (2^m m!)^{-1/2} \int H_m(u)\, G\, {\rm d}u$	same, $\tilde{G}_m^\pm$ (B27, B30)	—	same

A.1 Laguerre projection and the branches

With $x = v_\perp^2/v_{\rm th}^2$, CMM (eq. B8) and S09 (eq. 205) write

$$g = \frac{n_i}{\pi v_{\rm th}^2}\, e^{-x} \sum_{l\ge 0} L_l(x)\, G_l(v_\parallel), \qquad F_0 = \frac{n_i}{\pi v_{\rm th}^2}\, e^{-x} F_M(v_\parallel),$$

with L_l orthonormal with weight e^{-x} on $(0,\infty)$. Since $x = L_0 - L_1$ and $x^k \perp L_l$ for $l > k$, the source term of (3) projects only on L_0 and L_1, and (3) and (4) become

$$\frac{{\rm d}G_0}{{\rm d}t} + v_\parallel \nabla_\parallel \Big[G_0 + F_M\Big(\frac{Z}{\tau}\delta_n + \delta_B\Big)\Big] = 0, \qquad \frac{{\rm d}G_1}{{\rm d}t} + v_\parallel \nabla_\parallel (G_1 - F_M \delta_B) = 0,$$
$$\frac{{\rm d}G_l}{{\rm d}t} + v_\parallel \nabla_\parallel G_l = 0 \quad (l \ge 2),$$
$$\int G_0\, {\rm d}v_\parallel = \delta_n - \delta_B, \qquad \int (G_0 - G_1)\, {\rm d}v_\parallel = -\frac{Z}{\tau}\delta_n - 2\Big(1 + \frac{1}{\beta_i}\Big)\delta_B.$$

With $D = Z/\tau + 2(1+\beta_i^{-1})$, the distributions of CMM's eqs (B2)–(B3) (S09 eqs 176–177),

$$G_n = -\frac{1}{D}\Big[(G_0 - G_1) - 2\Big(1 + \frac{1}{\beta_i}\Big)G_0\Big], \qquad G_B = -\frac{1}{D}\Big[(G_0 - G_1) + \frac{Z}{\tau}G_0\Big],$$

have $\int G_n\, {\rm d}v_\parallel = \delta_n$ and $\int G_B\, {\rm d}v_\parallel = \delta_B$, invert to

$$G_0 = G_n - G_B, \qquad G_1 = \Big(1 + \frac{Z}{\tau}\Big)G_0 + D\, G_B, \tag{A1}$$

and obey S09's eqs (179)–(180),

$$\frac{\rm d}{{\rm d}t}\begin{pmatrix} G_n \\ G_B \end{pmatrix} + v_\parallel \nabla_\parallel \begin{pmatrix} G_n \\ G_B \end{pmatrix} = v_\parallel F_M\, \nabla_\parallel\, K \begin{pmatrix} \delta_n \\ \delta_B \end{pmatrix}, \qquad K = \frac{1}{D}\begin{pmatrix} -\frac{Z}{\tau}\Big(1 + \frac{2}{\beta_i}\Big) & -\frac{2}{\beta_i} \\ \frac{Z}{\tau}\Big(1 + \frac{Z}{\tau}\Big) & 2 + \frac{Z}{\tau} \end{pmatrix}.$$

If $\ell K = \Lambda^{-1}\ell$, then $G = \ell\cdot(G_n, G_B)$ has $\int G\, {\rm d}v_\parallel = \ell\cdot(\delta_n, \delta_B)$ and obeys (5). The two left eigenvectors give CMM's branches (B5)–(B6) (S09 eqs 183–184),

$$G^+ = G_B + \frac{1 + Z/\tau}{\sigma}\, G_n, \qquad G^- = G_n + \frac{2\tau}{\sigma Z \beta_i}\, G_B, \qquad \sigma = 1 + \frac{\tau}{Z} + \frac{1}{\beta_i} + \Delta,$$

and the eigenvalues are $1/\Lambda^\pm$ with $\Lambda^\pm$ and Δ of (6).

A.2 Moments

The map $(G_n, G_B) \mapsto (G^+, G^-)$ has determinant $-2\Delta/\sigma \neq 0$ and inverse

$$G_n = \frac{\sigma G^- - (2\tau/Z\beta_i)\, G^+}{2\Delta}, \qquad G_B = \frac{\sigma G^+ - (1 + Z/\tau)\, G^-}{2\Delta}, \tag{A2}$$

so that, with $M_0^\pm = \int G^\pm\, {\rm d}v_\parallel$,

$$\delta_n = \frac{\sigma M_0^- - (2\tau/Z\beta_i)\, M_0^+}{2\Delta}, \qquad \delta_B = \frac{\sigma M_0^+ - (1 + Z/\tau)\, M_0^-}{2\Delta}. \tag{A3}$$

These are one half of the right-hand sides of S09's eqs (202)–(203), as CMM found (footnote to their eq. B15). The parallel flow and pressure perturbations of the non-Alfvénic part $g_i = g + x\,\delta_B F_0$ of the ion distribution (CMM eq. 3.26) are

$$u_\parallel = \int v_\parallel G_0 \,\mathrm{d}v_\parallel, \qquad \frac{\delta p_\perp}{p} = -\frac{Z}{\tau}\delta_n - \frac{2}{\beta_i}\delta_B, \qquad \frac{\delta p_\parallel}{p} = \frac{2}{v_{\mathrm{th}}^2}\int v_\parallel^2 G_0 \,\mathrm{d}v_\parallel + \delta_B,$$

with $p = n_i T_i$. By (A1) and (A2) these are linear in $G^\pm$, and the G_l with $l \geq 2$ enter only moments $\int \mathrm{d}^3v\, v_\parallel^j v_\perp^{2k} g_i$ with $k \geq 2$.

A.3 Free energy

With $x = L_0 - L_1$, the orthonormality of the L_l and the field relations, CMM's compressive free energy (their eq. 3.31) becomes

$$\begin{aligned} W_{\mathrm{compr}} &= \frac{n_i T_i}{2}\Big\langle \frac{Z}{\tau}\delta_n^2 + \frac{2}{\beta_i}\delta_B^2 + \frac{1}{n_i}\int \mathrm{d}^3v\, \frac{g_i^2}{F_0}\Big\rangle \\ &= \frac{n_i T_i}{2}\Big\langle \int \frac{G_0^2 + G_1^2}{F_M}\,\mathrm{d}v_\parallel + \frac{Z}{\tau}\delta_n^2 - \frac{2Z}{\tau}\delta_n\delta_B - 2\Big(1+\frac{1}{\beta_i}\Big)\delta_B^2 + \sum_{l\geq 2}\int \frac{G_l^2}{F_M}\,\mathrm{d}v_\parallel\Big\rangle. \end{aligned}$$

The part with $l = 0, 1$ splits between the branches by the identities

$$G_0^2 + G_1^2 = c_+\,(G^+)^2 + c_-\,(G^-)^2, \qquad \frac{Z}{\tau}\delta_n^2 - \frac{2Z}{\tau}\delta_n\delta_B - 2\Big(1+\frac{1}{\beta_i}\Big)\delta_B^2 = -c_+\frac{(M_0^+)^2}{\Lambda^+} - c_-\frac{(M_0^-)^2}{\Lambda^-}, \tag{A4}$$

the first of which holds at each $v_\parallel$, with

$$c_+ = \Big[1 + \frac{1+\tau/Z}{\Delta}\Big](\Lambda^+)^2 > 0, \qquad c_- = \frac{Z^2}{2\tau^2}\Big[1 + \frac{1}{\Delta\beta_i}\Big](\Lambda^-)^2 > 0. \tag{A5}$$

This gives CMM's eq. (B15) with their eqs (B13)–(B14),

$$W_{\mathrm{compr}} = \tilde{W}_{\mathrm{compr}} + c_+ W_{\mathrm{compr}}^+ + c_- W_{\mathrm{compr}}^-, \qquad W_{\mathrm{compr}}^\pm = \frac{n_i T_i}{2}\Big\langle \int \frac{(G^\pm)^2}{F_M}\,\mathrm{d}v_\parallel - \frac{(M_0^\pm)^2}{\Lambda^\pm}\Big\rangle, \tag{A6}$$

with $\tilde{W}_{\mathrm{compr}}$ the part with $l \geq 2$. Each part is conserved (S09, §§6.2.3 and 6.2.5; CMM eq. B12) and positive definite (section 3).

The coefficients (A5) use Δ of (6) (S09 eq. 204) and are one quarter of those in S09's eq. (213), as CMM state. CMM's eq. (B16) prints this square root as $[(1+\tau/Z) + \beta_i^{-2}]^{1/2}$, which gives $\sqrt{3}$ instead of $\sqrt{5}$ at $\beta_i = \tau = Z = 1$. Their eqs (B6)–(B7) are correct, so the misprint affects only $c_\pm$ in (B15) and (B17) and our (A6), which then fails. For random distribution functions the relative error in W_{compr} with the printed square root is up to 7.2×10^{-2} over the parameters we tried (2.5×10^{-2} at $\beta_i = \tau = Z = 1$), and below 5×10^{-14} with the square restored.

B The slice transform and the spectral theory of the branch operator

B.1 Definitions and statement

Let $\mathcal{W}$ be the space of real functions $G(u)$ with $\langle G, G\rangle < \infty$, where $\langle f, g\rangle = \int f g\, F_M^{-1}\,\mathrm{d}u$, so that $\mathsf{Q} = F_M\, M_0$ is the orthogonal projector onto F_M. The free-energy product (12) is $[f, g] = \langle f, Pg\rangle$

with $P = I - \mathsf{Q}/\Lambda$, and with $G = M_0[G]\,F_M + G_\perp$, $[G,G] = \langle G_\perp, G_\perp\rangle + p_0\,M_0[G]^2$, which is positive definite, with a norm equivalent to that of $\langle\cdot,\cdot\rangle$, if and only if $\Lambda \notin [0,1]$. With U the multiplication by u,

$$L = UP, \tag{B1}$$

so $[f, Lg] = \langle Pf, UPg\rangle$ is symmetric in f and g. For $\operatorname{Im} z > 0$,

$$(z-L)^{-1}G = \frac{1}{z-u}\left[G - \frac{uF_M}{\Lambda\,\varepsilon(z)}\int \frac{G(v)}{z-v}\,\mathrm{d}v\right], \qquad \varepsilon(\zeta) = 1 - \frac{1+\zeta Z(\zeta)}{\Lambda}, \tag{B2}$$

where Z is the plasma dispersion function of Fried and Conte (1961), not the charge number; we write ε also for its continuation into the lower half plane, whose zeros are the Landau roots. On the real axis

$$\varepsilon(u+\mathrm{i}0) = \varepsilon_R(u) + \mathrm{i}\,\varepsilon_I(u), \qquad \varepsilon_R = 1 - \frac{1+u\operatorname{Re} Z(u)}{\Lambda}, \qquad \varepsilon_I = -\frac{\pi\,u\,F_M(u)}{\Lambda}, \tag{B3}$$

and $\mathsf{H}[f](u) = \pi^{-1}\mathrm{P}\!\int f(v)\,(v-u)^{-1}\,\mathrm{d}v$. By the argument of Penrose (1960), ε has no zeros in the closed upper half plane for $\Lambda \notin [0,1]$ (section B.5). The spectral density

$$\varrho(u) = \frac{F_M(u)}{|\varepsilon(u+\mathrm{i}0)|^2} \tag{B4}$$

is then positive, smooth and even, with $\int \varrho\,\mathrm{d}u = 1/p_0$. For real u the equation $(L-u)F = 0$ has the Case–Van Kampen solutions

$$F_u(v) = \frac{1}{\Lambda}\,\mathrm{P}\frac{vF_M(v)}{v-u} + \varepsilon_R(u)\,\delta(v-u), \qquad M_0[F_u] = 1, \qquad M_1[F_u] = u, \tag{B5}$$

and the slice transform is the projection on them in the free-energy product,

$$\mathcal{T}[G](u) = [F_u, G] = \frac{\varepsilon_R\,G - \varepsilon_I\,\mathsf{H}[G]}{F_M}(u), \qquad \mathcal{T}[LG](u) = u\,\mathcal{T}[G](u), \tag{B6}$$

of which the slice amplitude (13) is the unit-weight form, $a_u = \sqrt{\varrho(u)}\,\mathcal{T}[G](u)$; $\mathcal{A}$ denotes the map $G \mapsto a$.

Theorem 4. *Let $\Lambda \notin [0,1]$, the value $\Lambda = \infty$ included. Then L, with the domain $D(U)$ of U, is self-adjoint in $(\mathcal{W}, [\cdot,\cdot])$; it has no eigenvalues, and its spectrum is the real line, purely absolutely continuous and simple; and $\mathcal{A}$ is a unitary map from $(\mathcal{W}, [\cdot,\cdot])$ onto $L^2(\mathbb{R}, \mathrm{d}u)$ with $\mathcal{A}LG = u\,\mathcal{A}G$ for $G \in D(U)$. In particular (14) holds, $[G, \varphi(L)G] = \int \varphi(u)\,a_u^2\,\mathrm{d}u$ for every bounded Borel function φ, and*

$$G = \int \varrho(u)\,\mathcal{T}[G](u)\,F_u\,\mathrm{d}u = \varepsilon_R\,c + \varepsilon_I\,\mathsf{H}[c], \qquad c = \sqrt{\varrho}\;a. \tag{B7}$$

The rest of this appendix proves the theorem and the statements about the eigenvalues of L and the zeros of ε. For $0 < \Lambda < 1$, ε has one zero in the upper half plane, on the imaginary axis, L has the eigenvalues $\pm\mathrm{i}\gamma$, and the slice decomposition is incomplete.

B.2 Preliminaries

We work in the complexification of $\mathcal{W}$, with $\langle f, g\rangle = \int \bar{f}\, g\, F_M^{-1}\,\mathrm{d}u$. The set $\mathcal{D}_0$ of functions pF_M, with p a polynomial, is dense in $\mathcal{W}$ (it contains the orthonormal basis e_m), lies in $D(U)$, and is mapped into itself by U, P and L. Let $\mathcal{S}$ be the set of smooth f with $|f(v)| \le C(1+|v|)^k F_M(v)$ for some C and k; it contains $\mathcal{D}_0$ and the smooth functions with compact support. For $f \in \mathcal{S}$ the Cauchy integral

$$\chi_f(\zeta) = \frac{1}{\pi}\int \frac{f(v)}{v-\zeta}\,\mathrm{d}v \qquad (\operatorname{Im}\zeta > 0)$$

is analytic in the upper half plane, extends continuously to the real axis with $\chi_f(u+\mathrm{i}0) = \mathsf{H}[f](u) + \mathrm{i}f(u)$ (the Plemelj formula), and is $O(1/\zeta)$ as $|\zeta| \to \infty$ in the closed upper half plane. The following converse is used twice.

Lemma 1. *Let Y be analytic in the upper half plane, continuous up to the real axis, and $O(1/\zeta)$ as $|\zeta| \to \infty$ in the closed upper half plane. Then $Y = \chi_y$ with $y(u) = \operatorname{Im} Y(u+\mathrm{i}0)$.*

Proof. For $\operatorname{Im}\zeta > 0$, Cauchy's formula on a large half disc gives $Y(\zeta) = (2\pi\mathrm{i})^{-1}\int Y(v+\mathrm{i}0)\,(v-\zeta)^{-1}\,\mathrm{d}v$, and Cauchy's theorem for $\overline{Y(\bar\zeta)}$, analytic in the lower half plane with the same decay, gives $\int \overline{Y(v+\mathrm{i}0)}\,(v-\zeta)^{-1}\,\mathrm{d}v = 0$. Subtracting $(2\pi\mathrm{i})^{-1}$ times the second integral from the first gives $Y = \chi_y$. □

B.3 Self-adjointness

Let $\Lambda \notin [0,1]$, so that $p_0 > 0$. Since $P = (I-\mathsf{Q}) + p_0\mathsf{Q}$, the operator $P^{1/2} = I + (\sqrt{p_0}-1)\mathsf{Q}$ is bounded and symmetric, with bounded inverse, and $[f,g] = \langle P^{1/2}f, P^{1/2}g\rangle$, so $V : G \mapsto P^{1/2}G$ is unitary from $(\mathcal{W}, [\cdot,\cdot])$ onto $(\mathcal{W}, \langle\cdot,\cdot\rangle)$. Since $V - I$ has its range in the span of $F_M \in D(U)$, V maps $D(U)$ onto itself, and by (B1)

$$VLV^{-1} = P^{1/2}UP^{1/2} = U + \mathsf{K}, \qquad \mathsf{K} = (\sqrt{p_0}-1)(\mathsf{Q}U + U\mathsf{Q}), \tag{B8}$$

where $(\sqrt{p_0}-1)^2\mathsf{Q}U\mathsf{Q}$ is absent because $\mathsf{Q}U\mathsf{Q}G = M_1[F_M]\,M_0[G]\,F_M = 0$. The operator $\mathsf{K}G = (\sqrt{p_0}-1)\big[\langle uF_M, G\rangle\,F_M + \langle F_M, G\rangle\,uF_M\big]$ is symmetric, of rank two and bounded, since $\langle F_M, F_M\rangle = 1$ and $\langle uF_M, uF_M\rangle = 1/2$. By the Kato–Rellich theorem (Kato, 1966, chapter V), $U+\mathsf{K}$ is then self-adjoint on $D(U)$ in $(\mathcal{W}, \langle\cdot,\cdot\rangle)$, so $L = V^{-1}(U+\mathsf{K})V$ is self-adjoint on $D(U)$ in $(\mathcal{W}, [\cdot,\cdot])$, which is part (i) of theorem 4.

B.4 Eigenvalues

For every $\Lambda \neq 0$, a function $G \in D(U)$ is an eigenfunction of L with eigenvalue λ if and only if

$$(u-\lambda)\,G = \frac{uF_M}{\Lambda}\,M_0[G], \qquad G \neq 0.$$

If λ is not real, $G = M_0[G]\,uF_M/[\Lambda(u-\lambda)]$, and taking M_0 gives $M_0[G]\,\varepsilon(\lambda) = 0$ if $\operatorname{Im}\lambda > 0$ and $M_0[G]\,\overline{\varepsilon(\bar\lambda)} = 0$ if $\operatorname{Im}\lambda < 0$, because the integral in (B2) at a point of the lower half plane equals $\overline{\varepsilon(\bar\lambda)}$ (for the same reason the resolvent for $\operatorname{Im} z < 0$ is (B2) with $\overline{\varepsilon(\bar z)}$ in place of $\varepsilon(z)$). Since $M_0[G] = 0$ would give $G = 0$, a non-real λ is an eigenvalue if and only if λ or $\bar\lambda$ is a zero of ε in the upper half plane. For real $\lambda \neq 0$, $uF_M/(u-\lambda)$ is not square integrable near $u = \lambda$, so $M_0[G] = 0$ and $G = 0$. For $\lambda = 0$, $G = M_0[G]\,F_M/\Lambda$ requires $\Lambda = 1$.

B.5 Zeros of the dielectric function

The function ε is analytic in the upper half plane and continuous up to the real axis, where it takes the values (B3). Since $1+\zeta Z(\zeta) = -1/(2\zeta^2) + O(\zeta^{-4})$ uniformly in $\operatorname{Im}\zeta \geq 0$, $\varepsilon \to 1$ as $|\zeta| \to \infty$ there, so if ε has no real zeros, the number of its zeros in the upper half plane is the winding number about 0 of the closed curve $u \mapsto \varepsilon(u+\mathrm{i}0)$. A real zero requires $\varepsilon_I(u) = 0$, hence $u = 0$, and $\varepsilon_R(0) = p_0$ vanishes only at $\Lambda = 1$. For $\Lambda \neq 1$, $\varepsilon(u+\mathrm{i}0) = 1 - \mathcal{N}(u)/\Lambda$ with $\mathcal{N}(u) = 1 + uZ(u+\mathrm{i}0)$, and the winding number of ε about 0 equals that of $\mathcal{N}$ about the point Λ. The curve $\mathcal{N}$ has $\operatorname{Im}\mathcal{N}(u) = \pi u F_M(u)$, $\mathcal{N}(0) = 1$ and $\mathcal{N}(u) \to 0$ as $|u| \to \infty$: it leaves 0 at $u = -\infty$, runs through the lower half plane to the point 1, reached at $u = 0$, and returns to 0 through the upper half plane, meeting the real axis only at 0 and 1. It therefore winds once around every point of $(0,1)$ and around no real point outside $[0,1]$, so

$$\#\{\zeta : \varepsilon(\zeta) = 0,\ \operatorname{Im}\zeta \geq 0\} = \begin{cases} 0, & \Lambda < 0 \text{ or } \Lambda > 1, \\ 1, & 0 < \Lambda < 1, \end{cases} \tag{B9}$$

which is the criterion of Penrose (1960) for this dielectric function. By section B.4, L has no eigenvalues for $\Lambda \notin [0,1]$.

The plasma dispersion function continues to the entire function $Z(\zeta) = \mathrm{i}\sqrt{\pi}\, e^{-\zeta^2} \operatorname{erfc}(-\mathrm{i}\zeta)$, and the zeros of the continued ε in the lower half plane are the Landau roots. They are not eigenvalues of L, because in the lower half plane the resolvent contains $\overline{\varepsilon(\bar{\zeta})}$, which by (B9) has no zeros there for $\Lambda \notin [0,1]$.

B.6 Spectral measure

For $\operatorname{Im} z > 0$ and real $G \in \mathcal{W}$, let

$$C_G(z) = \int \frac{G(v)}{z-v}\,\mathrm{d}v, \qquad I_G(z) = \int \frac{G(v)^2}{(z-v)\,F_M(v)}\,\mathrm{d}v.$$

By (B2), $M_0[(z-L)^{-1}G] = C_G(z)/\varepsilon(z)$, and since $\int uG(u)\,(z-u)^{-1}\,\mathrm{d}u = zC_G(z) - M_0[G]$, the resolvent form is

$$R_G(z) \equiv [G, (z-L)^{-1}G] = I_G(z) - \frac{z}{\Lambda}\,\frac{C_G(z)^2}{\varepsilon(z)}.$$

It is the Stieltjes transform $\int (z-x)^{-1}\,\mathrm{d}\mu_G(x)$ of the spectral measure μ_G of L in the state G, which Stone's formula recovers:

$$\mu_G((u_1,u_2)) + \tfrac{1}{2}\mu_G(\{u_1\}) + \tfrac{1}{2}\mu_G(\{u_2\}) = -\frac{1}{\pi}\lim_{\eta\to 0+}\int_{u_1}^{u_2} \operatorname{Im} R_G(u+\mathrm{i}\eta)\,\mathrm{d}u. \tag{B10}$$

Let $G \in \mathcal{D}_0$ be real and $\Lambda \notin [0,1]$. Then $C_G = -\pi\chi_G$ and $I_G = -\pi\chi_{G^2/F_M}$ extend continuously to the real axis, with $C_G(u+\mathrm{i}0) = -\pi(\mathsf{H}[G] + \mathrm{i}G)(u)$ and $\operatorname{Im} I_G(u+\mathrm{i}0) = -\pi G(u)^2/F_M(u)$. By (B9), $1/\varepsilon$ is continuous on the closed upper half plane, so $R_G(u+\mathrm{i}\eta) \to R_G(u+\mathrm{i}0)$ as $\eta \to 0+$, uniformly for u in compact sets. With $\varepsilon_I = -\pi u F_M/\Lambda$, the imaginary part of the boundary value is a square:

$$-\frac{1}{\pi}\operatorname{Im} R_G(u+\mathrm{i}0) = \frac{G^2}{F_M} - \frac{\varepsilon_I}{F_M}\,\frac{2\varepsilon_R\, G\,\mathsf{H}[G] - \varepsilon_I(\mathsf{H}[G]^2 - G^2)}{|\varepsilon|^2} = \frac{\left(\varepsilon_R G - \varepsilon_I \mathsf{H}[G]\right)^2}{F_M\,|\varepsilon|^2} = \varrho\,\mathcal{T}[G]^2 = a_u^2.$$

By (B10) and the uniform convergence, μ_G has no atoms and $\mathrm{d}\mu_G = a_u^2\,\mathrm{d}u$.

Three consequences follow. First, the absolutely continuous subspace of L is closed and contains the dense set $\mathcal{D}_0$, so it is $\mathcal{W}$, and L has no eigenvalues and no singular continuous spectrum. Second, K in (B8) has rank two and is therefore of trace class, so by the Kato–Rosenblum theorem (Kato, 1966, chapter X) the absolutely continuous parts of $U + \mathsf{K} = VLV^{-1}$ and of U are unitarily equivalent; since both operators are purely absolutely continuous, L is unitarily equivalent to U, multiplication by u in $L^2(\mathbb{R}, \mathrm{d}u/F_M)$, whose spectrum is the real line with multiplicity one. This is part (ii) of theorem 4. Third, applying $\mathrm{d}\mu_G = a_u^2 \,\mathrm{d}u$ to $G+G'$ and $G-G'$ gives, for real $G, G' \in \mathcal{D}_0$ and every bounded Borel function φ,

$$[G', \varphi(L)G] = \int \varphi(u)\, a_u[G']\, a_u[G] \,\mathrm{d}u, \qquad a[G] = \mathcal{A}G, \tag{B11}$$

and by linearity the same holds for complex $G, G' \in \mathcal{D}_0$ with $a_u[G']$ replaced by its complex conjugate.

B.7 The slice transform

The formula (13), with H the Hilbert transform on $L^2(\mathbb{R}, \mathrm{d}u)$, defines a bounded operator $\mathcal{A}$ from $\mathcal{W}$ into $L^2(\mathbb{R}, \mathrm{d}u)$: for $\Lambda \notin [0,1]$, $|\varepsilon(u+\mathrm{i}0)|$ is bounded below, since it has no real zeros and tends to 1 at infinity, and since $|\varepsilon_R| \le |\varepsilon|$, the first term of (13) has norm at most $\langle G, G\rangle^{1/2}$. In the second, $\varepsilon_I \mathsf{H}[G]/(\sqrt{F_M}\,|\varepsilon|) = -\pi u \sqrt{F_M}\,\mathsf{H}[G]/(\Lambda|\varepsilon|)$, the factor $u\sqrt{F_M}/|\varepsilon|$ is bounded, H is an isometry, and $\int |G|^2 \,\mathrm{d}u \le \pi^{-1/2}\langle G, G\rangle$. By (B11) with $\varphi = 1$, $\mathcal{A}$ is isometric on the dense set $\mathcal{D}_0$, so it is an isometry of $(\mathcal{W}, [\cdot,\cdot])$ into $L^2(\mathbb{R}, \mathrm{d}u)$, and its range is closed.

The following argument is the counterpart for L of the invertibility proof of Morrison (2000). To show that the range of $\mathcal{A}$ is dense, let b be a smooth function with compact support, $c = \sqrt{\varrho}\, b$, and

$$G = \varepsilon_R c + \varepsilon_I \mathsf{H}[c] = \operatorname{Im}\bigl[\varepsilon(u+\mathrm{i}0)\,\chi_c(u+\mathrm{i}0)\bigr].$$

Since c has compact support, $\varepsilon_I = -\pi u F_M/\Lambda$ and $\mathsf{H}[c]$ is smooth and bounded, G belongs to $\mathcal{S}$ and to $\mathcal{W}$. The product $\varepsilon\chi_c$ is analytic in the upper half plane, continuous up to the real axis and $O(1/\zeta)$, so by lemma 1 it equals χ_G, and $\chi_c(u+\mathrm{i}0) = \chi_G(u+\mathrm{i}0)/\varepsilon(u+\mathrm{i}0)$. Taking imaginary parts, with the Plemelj formula for χ_G,

$$c = \frac{\operatorname{Im}\bigl[(\mathsf{H}[G] + \mathrm{i}G)\,\bar{\varepsilon}\bigr]}{|\varepsilon|^2} = \frac{\varepsilon_R G - \varepsilon_I \mathsf{H}[G]}{|\varepsilon|^2} = \varrho\,\mathcal{T}[G],$$

so that $\mathcal{A}G = b$. The smooth functions with compact support are dense in $L^2(\mathbb{R}, \mathrm{d}u)$ and the range of $\mathcal{A}$ is closed, so $\mathcal{A}$ is onto, with the inverse

$$\mathcal{A}^{-1}a = \varepsilon_R\sqrt{\varrho}\,a + \varepsilon_I\,\mathsf{H}\bigl[\sqrt{\varrho}\,a\bigr], \tag{B12}$$

which is (B7): since $|\varepsilon_R|\sqrt{\varrho/F_M} \le 1$ and $|\varepsilon_I|/\sqrt{F_M}$ and ϱ are bounded, the right-hand side is a bounded operator from $L^2(\mathbb{R}, \mathrm{d}u)$ into $\mathcal{W}$ that equals $\mathcal{A}^{-1}$ on a dense set. Conversely, for $G \in \mathcal{D}_0$, $Y = \chi_G/\varepsilon$ is analytic in the upper half plane, because ε has no zeros there, and $\operatorname{Im} Y(u+\mathrm{i}0) = \varrho\mathcal{T}[G] = c \in \mathcal{S}$; by lemma 1, $Y = \chi_c$, and $G = \operatorname{Im}\chi_G(u+\mathrm{i}0) = \operatorname{Im}[\varepsilon\, Y](u+\mathrm{i}0) = \varepsilon_R c + \varepsilon_I \mathsf{H}[c]$. The first direction does not use the zeros of ε in the upper half plane; the second uses their absence, and fails by the residue of Y at the zero $\mathrm{i}\gamma$ that ε has for $0 < \Lambda < 1$, where $1 - \sqrt{\pi}\,\gamma\, e^{\gamma^2}\operatorname{erfc}(\gamma) = \Lambda$ and L has the eigenvalues $\pm\mathrm{i}\gamma$ (section B.4); the slices are then not complete. The KRMHD branches never fall in this range, because $\Lambda^+ > 1$ and $\Lambda^- < 0$.

Part (iii) of theorem 4 follows. By (B11) with $\varphi(u) = (z-u)^{-1}$, $\operatorname{Im} z \ne 0$, $[G', (z-L)^{-1}G] = \int \overline{a_u[G']}\, a_u[G]\,(z-u)^{-1}\,\mathrm{d}u$ for $G, G' \in \mathcal{D}_0$, and by continuity for all $G, G' \in \mathcal{W}$. Since $\mathcal{A}$ is unitary,

the left-hand side is $\int \overline{a_u[G']}\, a_u[(z-L)^{-1}G]\,\mathrm{d}u$, and since $\mathcal{A}G'$ runs over $L^2(\mathbb{R},\mathrm{d}u)$, $\mathcal{A}(z-L)^{-1}G = (z-u)^{-1}\mathcal{A}G$. For $G \in D(U)$, applying this to $(z-L)G$ shows that $u\,\mathcal{A}G \in L^2(\mathbb{R},\mathrm{d}u)$ and $\mathcal{A}LG = u\,\mathcal{A}G$. The relations (14) are then the functional calculus of multiplication by u.

By (B5), $\int c(u)\,F_u(v)\,\mathrm{d}u = \varepsilon_R(v)\,c(v) + \varepsilon_I(v)\,\mathsf{H}[c](v)$, so (B12) is the completeness relation $G = \int \varrho\,\mathcal{T}[G]\,F_u\,\mathrm{d}u$ of the modes, and $\varrho\,\mathcal{T}[\mathcal{A}^{-1}a] = \sqrt{\varrho}\,a$ is the orthogonality relation $[F_u, F_{u'}] = \delta(u-u')/\varrho(u)$, in the sense of distributions.

C The truncated hierarchy

C.1 Weights, closures and eigenvectors

Let J be the ladder matrix of the zero closure at $\Lambda = \infty$, the symmetric Jacobi matrix of the h_m, and J_c that of the copy closure, which for $M \geq 2$ differs from J only in $(J_c)_{M,M-1} = \sqrt{M/2} + \sqrt{(M+1)/2}$. A finite Λ changes only $A_{10} = (1-1/\Lambda)\,J_{10}$, and column 0 of J and of J_c has the single non-zero entry $J_{10} = 1/\sqrt{2}$. Hence, with $P_\Lambda = \mathrm{diag}(1-1/\Lambda, 1, \ldots, 1)$,

$$A = J\,P_\Lambda \quad \text{(zero closure)}, \qquad A = J_c\,P_\Lambda \quad \text{(copy closure, } M \geq 2\text{)}, \tag{C1}$$

the Hermite form of $L = UP$ (B1). For the zero closure $\mathsf{P} = P_\Lambda$, and $\mathsf{P}A = P_\Lambda J P_\Lambda$ is symmetric. For the copy closure $D_c = \mathrm{diag}(1, \ldots, 1, p_M)$ satisfies $D_c J_c = J$, because $p_M (J_c)_{M,M-1} = \sqrt{M/2}$, so that with $\mathsf{P} = D_c P_\Lambda$ again $\mathsf{P}A = P_\Lambda J P_\Lambda$.

Since $LF_u = uF_u$ and the untruncated ladder matrix is the matrix of L in the Hermite basis, the Hermite coefficients $r_m(u)$ of F_u satisfy $\sum_n A_{mn} r_n = u\,r_m$ row by row, with $r_0 = M_0[F_u] = 1$:

$$r_1 = \sqrt{2}\,u, \qquad r_2 = u\,r_1 - \frac{1-1/\Lambda}{\sqrt{2}}\,r_0, \qquad \sqrt{\frac{m+1}{2}}\,r_{m+1} = u\,r_m - \sqrt{\frac{m}{2}}\,r_{m-1} \quad (m \geq 2). \tag{C2}$$

Thus r_m is a polynomial of degree m with $r_m(-u) = (-1)^m r_m(u)$, and $r_m = h_m$ at $\Lambda = \infty$. Rows 0 to $M-1$ of the truncated matrix coincide with those of the infinite matrix for both closures, and row M is the condition on the eigenvalues,

$$r_{M+1}(u_j) = 0 \quad \text{(zero closure)}, \qquad r_{M+1}(u_j) = r_{M-1}(u_j) \quad \text{(copy closure)}, \tag{C3}$$

the second from row M of the copy closure, $u\,r_M = [\sqrt{M/2} + \sqrt{(M+1)/2}\,]\,r_{M-1}$, and the recurrence at $m = M$.

Proof of theorem 2. (i) A is similar to the symmetric tridiagonal matrix $\mathsf{P}^{1/2} A\,\mathsf{P}^{-1/2}$, whose off-diagonal entries are non-zero because $A_{01}A_{10} = (1-1/\Lambda)/2 > 0$, so its eigenvalues are real and simple and the vectors $\mathsf{P}^{1/2}\boldsymbol{r}_j$ are orthogonal. By (C2) the vector $(r_0(u), \ldots, r_M(u))^{\mathsf{T}}$ satisfies rows 0 to $M-1$ of $A\boldsymbol{r} = u\boldsymbol{r}$ for every u, (C3) fixes the u_j, and $\mathsf{P}A = A^{\mathsf{T}}\mathsf{P}$ gives $\boldsymbol{w}_j^{\mathsf{T}} A = u_j \boldsymbol{w}_j^{\mathsf{T}}$. (ii) The combination $\boldsymbol{w}_j^{\mathsf{T}} g$ commutes with the brackets, with ∂_z and with the linear operations, and $\boldsymbol{w}_j^{\mathsf{T}} A = u_j \boldsymbol{w}_j^{\mathsf{T}}$ turns $s\nabla_{\parallel}(Ag)$ into $s u_j \nabla_{\parallel}(\boldsymbol{w}_j^{\mathsf{T}} g)$; the converse follows from (iii). (iii) By (i), $R^{\mathsf{T}}\mathsf{P}R = I$, so $g = Ra$ and $\sum_m p_m |g_m|^2 = a^\dagger R^{\mathsf{T}}\mathsf{P}R\,a = \sum_j |a_j|^2$. □

C.2 Gauss quadrature

We prove proposition 2. For the basis functions e_m, $\mathcal{T}[e_m](u) = [F_u, e_m] = p_m\,r_m(u)$. (i) By theorem 4 the slice transform is unitary, and the slice amplitude of e_m is $\sqrt{\varrho}\,p_m r_m$, so $p_m \delta_{mn} = [e_m, e_n] = \int \varrho\,p_m r_m\,p_n r_n\,\mathrm{d}u$. (ii) The polynomials $\sqrt{p_m}\,r_m$ are therefore orthonormal for $\varrho\,\mathrm{d}u$, with the Jacobi matrix $\mathsf{P}^{1/2} A\,\mathsf{P}^{-1/2}$ by (C2), and the zero closure keeps its first $M+1$ rows and columns.

By the theorem of Golub and Welsch (1969), the eigenvalues of this truncated matrix are the zeros of r_{M+1} and the nodes of the Gauss rule, and the weights are $\mu_0 O_{0j}^2$, with O its orthonormal eigenvectors and $\mu_0 = \int \varrho \, \mathrm{d}u = 1/p_0$. Since $O_{mj} = \sqrt{p_m}\, R_{mj}$, the weights are $R_{0j}^2 = 1/n_j$, and exactness to degree $2M+1$ is Gauss's theorem. (iii) By definition $\boldsymbol{r}_j$ holds the Hermite coefficients of F_{u_j}, and $\mathcal{T}[G](u_j) = \sum_m g_m p_m r_m(u_j) = \boldsymbol{w}_j^{\mathsf{T}} g = \sqrt{n_j}\, a_j$.

C.3 The largest phase velocity

For the zero closure $A = J + \delta A$ exactly, with $\delta A = -(\Lambda\sqrt{2})^{-1} \boldsymbol{e}_1 \boldsymbol{e}_0^{\mathsf{T}}$ and $\boldsymbol{e}_m$ the mth unit vector. The normalised eigenvectors of J have components $(\lambda_j^\infty)^{1/2} h_m(x_j)$, and $h_0 = 1$, $h_1(x) = \sqrt{2}\, x$, so to first order $u_j = x_j - \lambda_j^\infty x_j/\Lambda + O(\Lambda^{-2})$. The weight λ_M^∞ is of order $e^{-x_M^2}$ up to an algebraic factor, with $x_M^2 = 2M + 3 - 2^{2/3}|\alpha_1|(2M+3)^{1/3} + \ldots$, so the shift of $u_{\max}$ is about 10^{-49} at $M = 64$. Up to such terms, $u_{\max}$ is the largest zero of ψ_{M+1} for the zero closure and the largest root of $\psi_{M+1} = \psi_{M-1}$ for the copy closure.

The function ψ_{M+1} solves $\psi'' + (q^2 - x^2)\psi = 0$ with $q = q_{M+1}$. With $x = q + (2q)^{-1/3}\xi$ and $\sigma = (2q)^{-4/3}$, the uniform Airy (Langer) approximation $\psi \propto (\chi')^{-1/2}\mathrm{Ai}(\chi)$, with $\chi = \xi + \sigma\xi^2/5 + O(\sigma^2)$, places the largest zero at $\chi = \alpha_1$, which gives the first line of (28), the classical expansion of Szegő (1975, theorem 6.32) and Olver (1959) (DLMF, 2026, eq. 12.11.9). For the copy closure, $x\psi_M = \sqrt{M/2}\,\psi_{M-1} + \sqrt{(M+1)/2}\,\psi_{M+1}$ and $\psi_M' = \sqrt{M/2}\,\psi_{M-1} - \sqrt{(M+1)/2}\,\psi_{M+1}$ turn row M, $\psi_{M+1} = \psi_{M-1}$, into the Robin condition $\psi_M' + x(\sqrt{M}+\sqrt{M+1})^{-2}\psi_M = 0$ on ψ_M, which solves the same equation with $q = q_M$ and to leading order reads $\psi_\xi + \sigma\psi = 0$. The same approximation turns this condition into

$$\mathrm{Ai}'(\chi) + \tfrac{4}{5}\,\sigma\,\mathrm{Ai}(\chi) = 0, \qquad \text{so that} \qquad \chi = \alpha_1' - \frac{4\sigma}{5\alpha_1'} + O(\sigma^2),$$

and $\xi = \chi - \sigma\chi^2/5$ gives the second line of (28); $u_{\max}$ up to $M = 16384$ confirms both lines.

D Numerical methods

D.1 Energy accounting

Each step of the simulations of section 7 consists of GANDALF's Lawson-RK4 step with dissipation and hypercollisions, followed by the forcing increment. The hyper-resistivity and the hypercollisions are linear and diagonal in Fourier and Hermite space, so the free energy they remove from each moment during a step follows from the post-step fields and their damping factors, and the energy before dissipation is reconstructed exactly. The difference between that energy and the energy at the start of the step is the change due to the ideal part of the step, which the exact dynamics conserves; summed over the window it is the time-stepping error of table 1. The injected energy of each step is the change across the forcing increment. This gives $D_\perp(m)$, $D_\nu(m)$ and the injection rate in (21) per moment without finite differences in time, and the field-aligned flux is computed from (20) with GANDALF's dealiased bracket.

D.2 Field-line tracing

Field lines of $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}} \times \nabla_\perp \Psi$ are integrated in z with the classical fourth-order Runge–Kutta method and step $\Delta z = L/128$, with $\nabla_\perp \Psi$ interpolated by cubic B-splines on a grid refined by a factor of two by zero padding. We start 4096 lines at random points of the plane $z = 0$ and follow each over four box lengths. Along each line we sample the field of interest and compute its

spectrum in the arc length, which equals z to first order in $b_\perp$. The mean field-aligned wavenumber κ_1 is the first moment of $|\kappa|$ of the spectrum averaged over lines, and we average it over the nine snapshots. For the injected pattern we draw four fresh realisations of the density forcing with the run's generator and trace them on the same lines; their field-aligned spectrum, averaged over lines, draws and snapshots, is $P(\kappa)$.

D.3 The two-channel model

For one field-aligned wavenumber κ the model is the forced linear hierarchy

$$\frac{\mathrm{d}g}{\mathrm{d}t} = -\mathsf{L}g + \xi\,\boldsymbol{e}_0, \qquad \mathsf{L} = \mathrm{i}s\kappa A + \mathrm{diag}\Big(\frac{1}{2T} + \nu\Big(\frac{m}{M}\Big)^n[m \geq 2]\Big), \tag{D1}$$

where A is the truncated ladder matrix of the run, $\boldsymbol{e}_0$ the unit vector of g_0 and ξ white noise of intensity q. The damping $1/(2T)$ of every amplitude removes free energy from every moment at the rate $1/T$. The stationary covariance $\mathsf{C} = \langle gg^\dagger\rangle$ solves the Lyapunov equation $\mathsf{LC} + \mathsf{CL}^\dagger = q\,\boldsymbol{e}_0\boldsymbol{e}_0^\mathsf{T}$. Then $W(m) = p_m\mathsf{C}_{mm}$, the flux is $\Pi_{m+1/2} = -2s\kappa\,p_m A_{m,m+1}\,\mathrm{Im}\,\mathsf{C}_{m+1,m}$, the injection rate is $p_0 q$, and the slice covariance is $\mathsf{R}^\mathsf{T}\mathsf{P}\,\mathrm{Re}\,\mathsf{C}\,\mathsf{PR}$ with R the matrix of theorem 2. The model spectrum is the sum of these over the measured $P(\kappa)$, normalised so that the total injection rate equals the measured ε. Free energy injected at $\kappa = 0$ does not phase mix and is removed at $m = 0$ at the rate $1/T$. At large m and without hypercollisions, the sum reproduces (42).

D.4 Code and data

The configuration files, the run script and the analysis scripts accompany the paper in the folders `configs/`, `scripts/` and `analysis/`; the scripts read only saved data and regenerate every number and figure of section 7. The self-checking scripts behind sections 3 to 5 are in `derivations/`. [*To do:* data availability statement]

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